

Review Article

AI revolutionizing industries worldwide: A comprehensive overview of its diverse applications

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ABSTRACT

Artificial Intelligence (AI) technology's rapid advancement has significantly changed various industries' operations. This comprehensive review paper aims to provide readers with a deep understanding of AI's applications & implementations, workings, and potential impacts across different sectors. It also discusses its future, threats, and integration into new policy. Through extensive research on more than 200 research and many other sources, the authors have made every effort to present an accurate overview of the numerous applications of AI nowadays in industries such as agriculture, education, autonomous systems, healthcare, finance, entertainment, transportation, military, manufacturing, and more. The paper explores various AI technologies, including machine learning, deep learning, robotics, big data, the Internet of Things, natural language processing, image processing, object detection, virtual reality, augmented reality, speech recognition, and computer vision. It provides real-world examples of their applications and implementations. Moreover, it highlights and evaluates the future potential, challenges, and limitations associated with the widespread use of AI. Our study incorporates the latest research to offer a comprehensive and nuanced understanding of AI's potential benefits and challenges. This data-driven review case study highlights the immense potential of AI technology and addresses the ethical, societal, and economic considerations related to its implementation.

1. Introduction

We are in the fourth industrial revolution, where businesses and industries focus more on automation and machinery for efficient manufacturing and workforce optimization. Industrial Revolution 4.0 is turning the whole world upside down, replacing humans in more than several sectors by establishing new ways of doing things (in more optimizable and efficient routines), including how industry manufacturers transform goods, produce, follow the strategies to repair and install new production line and how other workforces in their relevant sectors implements plans, enhance, and distribute everything-all of these are part of Industrial Revolution 4.0 [272,276].

The foundations of Industry 4.0 are AI (Artificial Intelligence), Big Data, IoT (Internet of Things), Sensors, CPS (Cyber-Physical Systems, i.e. robots), and Blockchain [1]. So, in overall summary, cutting-edge techs like Cloud computing, Augmented Reality (AR), Virtual Reality (VR), Additive manufacturing (3D printing), Smart manufacturing, Advanced robotics, edge computing, Data analytics, Cyber-physical systems, cybersecurity solutions, Smart supply chain management, Image

processing, ML-based Object detection technology, Nanotechnology, Quantum computing, Cognitive computing, Digital twins, Natural language processing (NLP), and Advanced analytics, and every other thing that is relevant to automation in industries, businesses, and other sectors to reduce cost, time and labor-all are driven by Industry 4.0 which is eventually giving rise to another Industrial revolution, Industry 5.0. Previously, three more industrial revolutions before the Fourth Industrial Revolution of 2016 shaped the current function of the world and all the industrial sectors.

Fig. 1 highlights what sectors have been revolutionized over the years and based on what factors. Following the three is the fourth one that we are currently living in.

While the aim of Industry 4.0 was to reduce the human workforce to drive optimization in industries, the goal of Industry 5.0 is to open a phase where humans work alongside advanced technology and A.I.-powered robots to enhance workplace processes. This is coupled with a more human-centric focus, increased resilience, and improved focus on sustainability.

The emergence of Industry 5.0 marks a significant departure from

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the principles of Industry 4.0, extending beyond manufacturing to encompass all facets of business and organizational strategies. It builds upon the technological advancements of the fourth industrial revolution, including artificial intelligence, automation, big data analytics, the Internet of Things (IoT), machine learning, robotics, smart systems, and virtualization. Unlike its predecessor, Industry 5.0 prioritizes societal value and wellbeing over mere economic gains, as articulated by the European Union. Central to this shift is a focus on placing the welfare of workers at the forefront of the production process and leveraging new technologies to foster prosperity while respecting planetary constraints. This represents a fundamental reorientation towards a broader conception of value creation that transcends traditional notions of profit-driven growth.

The elaborate discussion of how Industry 4.0 changed the world, what ushered the dawn of Industry 5.0, and what they bring to the world is a vast study case of another domain. In this review paper, the authors offer in-depth, data-driven researched insights on what AI is, how the rapid advancement of AI technology has revolutionized the way industries operate, resulting in increased efficiency, productivity, and innovation, the history of AI, its vast implementation in different sectors in the last 25 years, and what is its future. Finally, it answers the burning question, 'Is it becoming a curse or a blessing, a tool for progress, or a source of concern?' prompting deeper reflection on the nuanced implications and potential outcomes of these transformative technological advancements nowadays.

By exploring the history of AI and its current applications in industries such as healthcare, finance, agriculture, marketing, and manufacturing, this research paper aims to shed light on AI's potential to transform how we live and work and further futuristic applications, and threats that can be created by AI while highlighting the challenges and opportunities that come with the adoption of this technology.

2. Overview of artificial intelligence (AI)

Artificial Intelligence (AI) encompasses machine-displayed intelligence that emulates human cognitive functions, such as problem-solving and decision-making. Combining Machine Learning and Deep Learning techniques, AI models are trained on extensive data sets to make intelligent decisions autonomously.

Fig. 2 demonstrates the simplest form of AI working. AI systems simulate human intelligence by analyzing data, recognizing patterns, and understanding natural language. Knowledge matching and explaining data through cognitive computing is the central part of AI functioning, as illustrated in the figure. They aim to replicate human cognitive abilities, including learning and perception, without direct human intervention. The global AI software market is projected to reach \$126 billion by 2025, with a 270 % increase in enterprise adoption over the past four years. AI is expected to power 95 % of customer interactions by 2025, with the market growing approximately 54 % year-on-year, reaching \$22.6 billion in size [3].

AI has been one of the best gifts in today's modern world. The capacity of a machine or computer system to imitate human intelligence in domains where human knowledge is typically required is referred to as

artificial intelligence or AI for short. Systems that use artificial intelligence analyze data using mathematical algorithms and statistical models, which allows them to improve over time [5]. The end goal of research into artificial intelligence is to develop intelligent machines that can see their surroundings, reason, learn, and act independently to accomplish challenging tasks. It has improved the world, the most significant difference between the twenty-first and previous centuries. The world has seen four industrial revolutions.

AI-enhanced technologies are continually being created to improve various human needs, including safety, security, entertainment, and many other areas. But before we reveal more about AI, let's break down what it is and what the idea of AI is [6]. It is the branch of computer science that enacts the ability of a machine or computer program to act like a human intellect or replicate human intelligence. The aim and objective of AI are to create intelligent machines called computed robots for various needs. They have their particular traits in the making [7]:

- Comprehension or proficiency in specific knowledge.
- Reasoning ability.
- Feature Engineering.
- Pattern recognition.
- Analysis and problem-solving.
- Perception.
- Data collecting and learning.
- Planning.

To date, these are the traits of AI, as illustrated in Fig. 3. Some even classify that trait into four categories [8]. AI personalities endow machines with distinct characteristics, facilitating more human-like user interactions. These personalities can be broadly categorized into four types: the Investigator, the Entertainer, the Helper, and the Challenger [9].

AI has been everywhere that people or users may not even know about, only because they didn't realize it or understand the idea of AI. It has been used in many aspects of our lives that we cannot move without its help. It is used by personal assistants like Amazon's Alexa, Apple's Siri, Microsoft's Cortana, and Google Assistant to comprehend what you say and carry out your orders. AI is also used by online entertainment platforms like Spotify and Netflix to predict every user's preference to make song and movie recommendations. To deliver tailored adverts, other services like Google, Facebook, Amazon, and eBay evaluate your online actions. So, it's clear that AI's presence is at every corner, from our daily life to entertainment, career, or daily servicing life. AI is also used in manufacturing, industrial automation, processing, finance, data collection and analysis, graphing, healthcare, agriculture, decision-making, head counting, shift accumulating, driverless cars, and many others [10].

Researchers from Google Health and Imperial College London created an algorithm that analyzed mammograms for breast cancer diagnosis better than six human radiologists. Groupe Renault is working with Google Cloud to combine AI and machine learning skills with automotive industry knowledge to boost productivity, raise production standards, and lower carbon emissions. AI is used by driverless

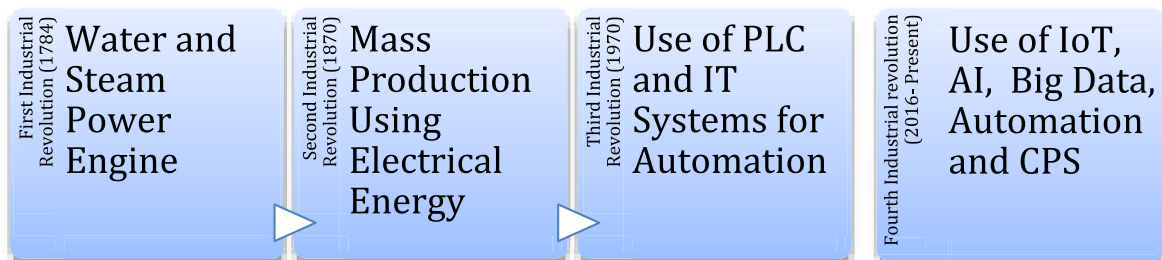


Fig. 1. Four Industrial Revolutions and their foundations [2].

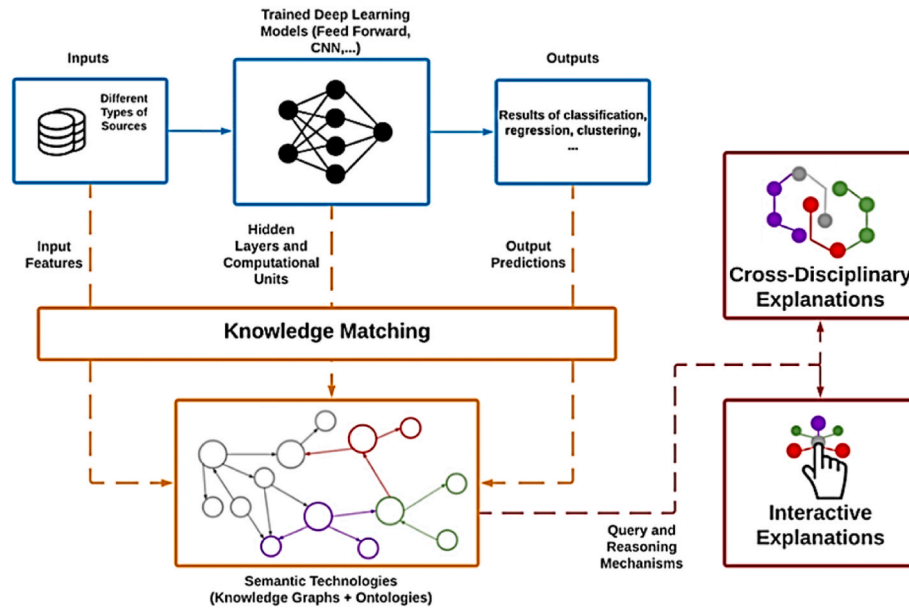


Fig. 2. Working principle of AI [4].

2.1. History of AI

AI remained a purely scientific field for many years. Following the evolutionary trajectory of their pertinent academic and scientific domains, such as mathematics, engineering physics, neurology, business study, computer science & programming, psychology, and others, universities and research laboratories developed their essential knowledge components [12]. Back in the 1940s, during World War II- for the first time, AI was traced when a British mathematician and computer scientist named Alan Turing came up with the idea and created the Bombe, a code-breaking device as shown in Fig. 2 that decoded German Enigma-encrypted signals in Bletchley Park, United Kingdom [13].

The Imitation Game, a 2014 Hollywood film, did a fantastic job portraying this historical era. The war is said to have been cut by more than two years because of Turing's efforts illustrated in Fig. 4, which also helped the Allies beat the Nazis and saved the lives of more than 14 million people [15].

The front panel of the bombe featured 104 electrical contacts arranged in four concentric circles around each shaft, forming a template of 312 contacts, as illustrated in Fig. 5. Wiring connected these contacts in a specific manner, allowing for the creation of double-ended

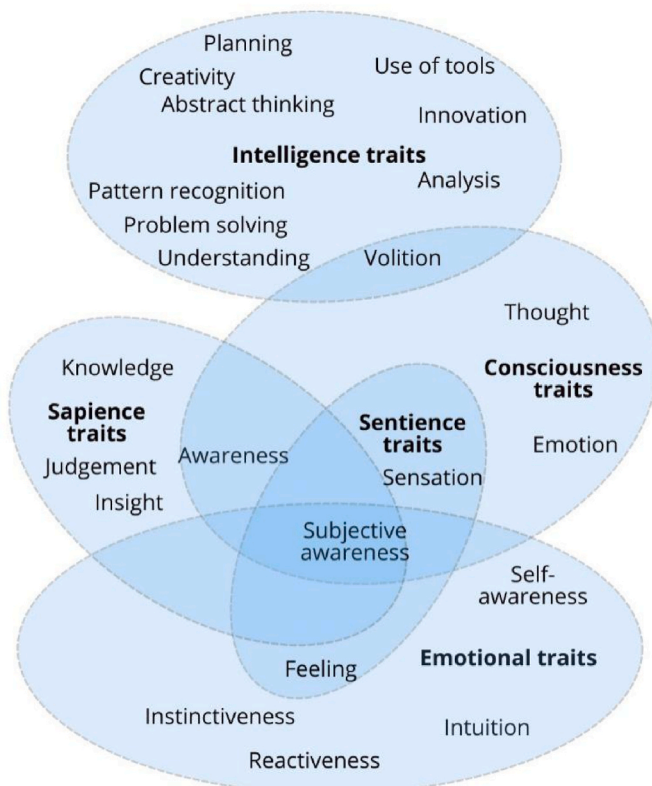


Fig. 3. The traits of AI.

automobiles to recognize roadways, pedestrians, and traffic signs. AI is used in finance to identify fraud and forecast future growth. AI is also used in agriculture to grow healthier crops, control pests, monitor soil and growth conditions, and do other things. AI can even impact our occupations. In the next 20 years, 35 % of today's occupations will vanish, according to the BBC. It has a broader way of utilization in the modern workplace worldwide, but before discussing them briefly, let's get along with the history of AI first [11].

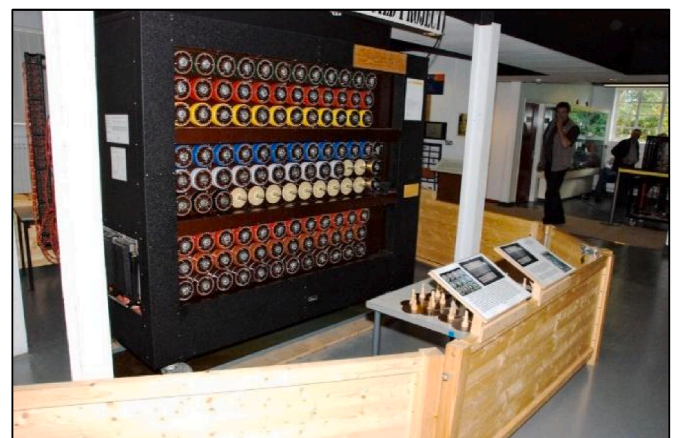


Fig. 4. The Bombe machine used in World War II (First ever courtesy of AI) [14].

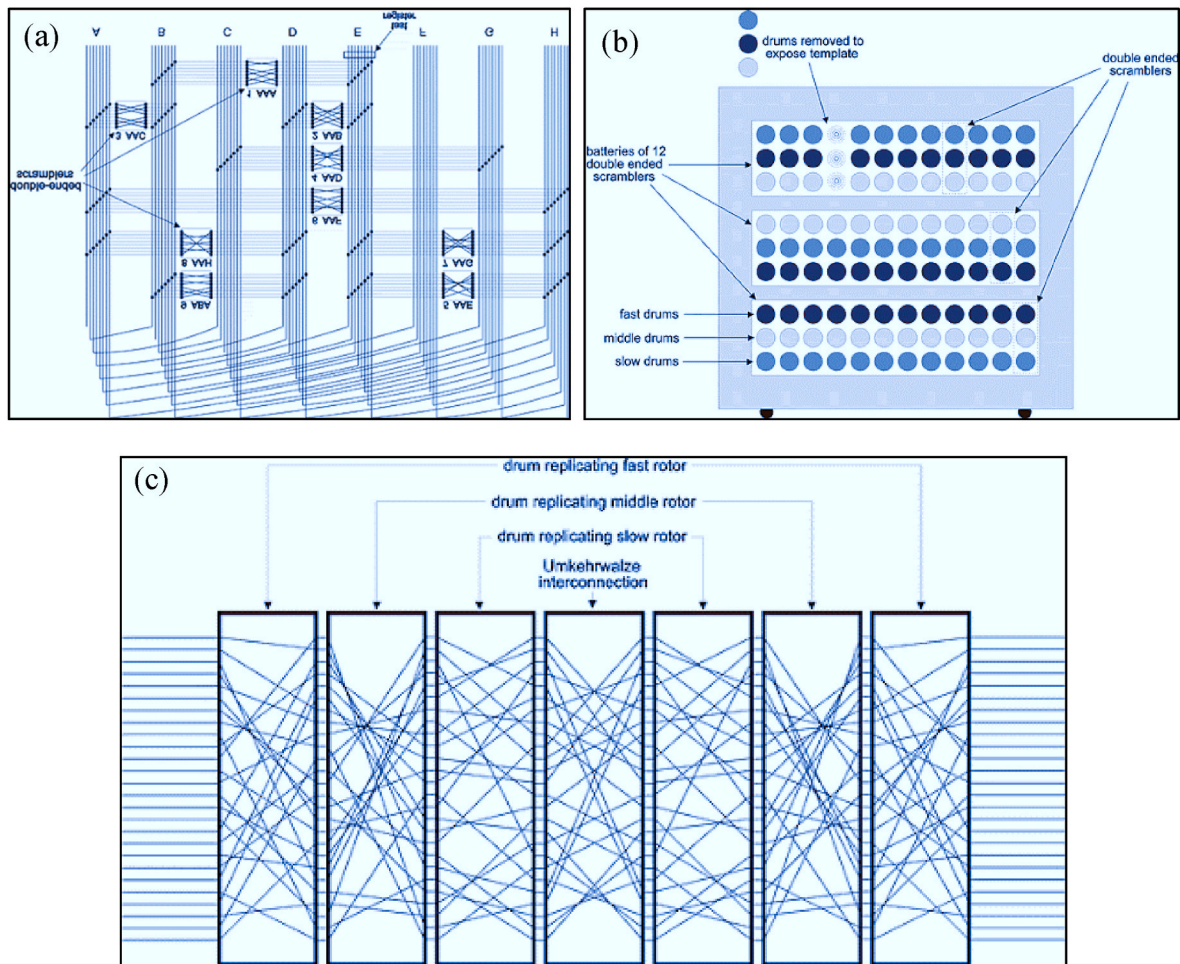


Fig. 5. The internal design of the Bombe machine [16].

scramblers. Each drum on the bombe replicated an Enigma rotor and had two sets of internal wiring to match the Enigma's internal wiring. These drums, colored according to their corresponding Enigma rotor, facilitated the transformation process. Fig. 3 (a & b) depicts the front panel of the bombe, while Fig. 5(c) illustrates the electrical pathways through a double-ended scrambler. Each battery of scramblers within the bombe was electrically isolated but mechanically connected, with an electric motor ensuring precise synchronization of the shafts and drums.

While employed, Turing wrote a paper for the journal *Mind* titled "Computing Machinery and Intelligence" in October 1950 (Oxford University Press). He conducted an experiment later known as the famous Turing test. The imitation game, a three-person activity, is frequently used to illustrate the Turing test. That game has three players: A human questioner, a Human respondent, and a computer respondent. The interrogator in the competition seeks to identify which of the players is a machine and which is a person. The interrogator can only make the determination using the answers to written questions. Since then, this specific machine has been used to determine if a machine's intellect is comparable to a human. The test was called the 'Turing test' per the scientist's name. The Turing test hasn't been passed by a computer yet. From that onward, the research for the term 'AI' took six years to be disciplined [17].

But the initiator of the AI revolution was much later. The Dartmouth Summer Research Project on Artificial Intelligence (DSRPAI) 1956 marked the beginning of AI research led by John McCarthy and Marvin Minsky. At this conference, Allen Newell, Cliff Shaw, and Herbert Simon introduced the Logic Theorist, the first AI program funded by the RAND Corporation. While the conference fell short of expectations due to a lack

of consensus on standard methods, it coined the term "artificial intelligence" and inspired confidence in the feasibility of AI. Despite initial challenges, the event catalyzed two decades of subsequent AI research and development [19].

1956 to 1974 is considered the golden period of AI. In 1966, researchers focused on algorithm development for solving mathematical problems. Joseph Weizenbaum introduced the first chatbot, ELIZA. Then, in 1972, Japan unveiled WABOT-1, the first intelligent humanoid robot [20]. The base of AI, Neural Networks (NN), was initiated in the 1950s–1970s. During this time, neural networks (termed artificial neural networks, in short ANNs) came into the foreplay of AI. It could mimic human biological neural networks as they were created based on human brains. Three layers typically make up a NN: the input layer, the hidden layer, and the output layer [13].

Fig. 6 dictates the primary level NN developed in the 80s. Things began to change between the 1980s and the 1990s when microprocessors, digital gadgets, networked computers, and the Internet were introduced. While convolutional neural networks (CNNs) originated in the 1980s, their significant breakthrough came in the 2000s, notably with the first GPU implementation described by K. Chellapilla et al., in 2006. This breakthrough was facilitated by the realization that standard neural networks could be substantially accelerated on graphics processing units (GPUs), as demonstrated by K. S. Oh and K. Jung in 2004. AI garnered growing business interest and heavy funding [13].

In 1980, following an AI winter period, the resurgence of AI was marked by the emergence of Expert Systems, which aimed to replicate the decision-making capabilities of human experts. The same year witnessed the inaugural national conference of the American Association of

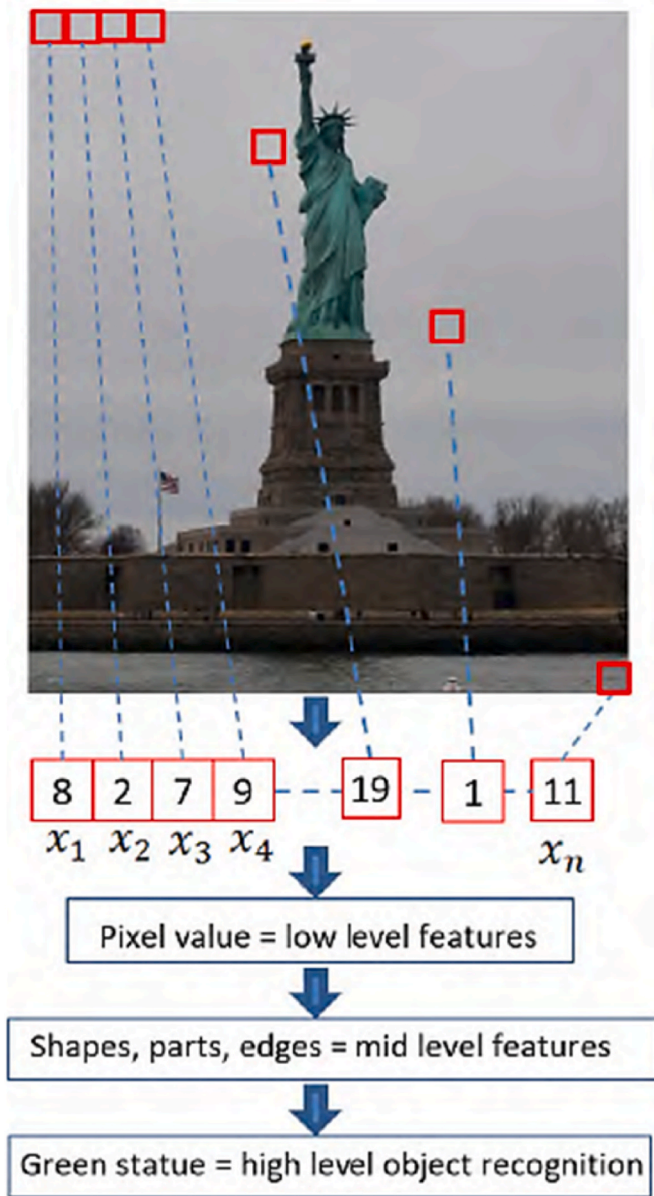


Fig. 6. Primary level of Image Processing architecture by CNN [18].

Artificial Intelligence at Stanford University.

However, the field encountered another setback during the second AI winter, from 1987 to 1993. During this period, investors and government agencies scaled back funding for AI research due to high costs and limited efficiency in producing tangible results; despite the cost-effectiveness of expert systems like XCON, the lack of significant breakthroughs led to a decline in support for AI initiatives.

From the 1990s to the 2010s, the idea of machine learning (ML) in the AI sector flourished and was nourished by many researchers and AI enthusiasts. This period is also considered the 'Emergence of Intelligent Systems' era. However, some believe this era to be from 1993 to 2011.

In 1997, IBM's Deep Blue defeated world chess champion Gary Kasparov, marking a landmark achievement in AI. By 2002, AI entered homes by introducing Roomba, the robotic vacuum cleaner. In 2006, AI expanded its presence in business as companies like Facebook, Twitter, and Netflix started integrating AI technologies for various applications.

Afterward, to date (2010 to present date), AI was broadly appreciated and taken for granted worldwide while Deep Learning (DL) was developed. DL is a particular kind of neural network with multiple

hidden layers. This conceivable is only with increased computing power, particularly from graphical processing units (GPUs) and improved algorithms. A subset of ML is DL [21]. On a sizable dataset, DL has so far outperformed many other methods. DL, like CNN, ANN, RNN, LSTM, and others, are advanced examples of the modern world. CNN is one of the most used and effective architectures (Fig. 4) in industrial and other relevant image-processing sectors. Their ability to automatically learn hierarchical features from raw pixel data makes them highly suitable for analyzing and processing visual information. CNNs have achieved state-of-the-art results in benchmark datasets and have been extensively used in industrial sectors such as manufacturing, quality control, medical imaging, and autonomous vehicles.

In 2011, IBM's Watson won Jeopardy, showcasing its natural language understanding. Google introduced "Google Now" in 2012 for predictive information. In 2014, the chatbot "Eugene Goostman" passed the Turing test. IBM's "Project Debater" engaged in complex debates four years later. Also, in 2018, Google's "Duplex" demonstrated human-like conversational abilities.

As shown throughout the ages, the stages of AI, per Fig. 4, is the detailed history of AI, while the history may be dispatched from its original history as the idea is often confused with data science, data mining, and big data. Yes, they are related to AI, but they are not the same thing as AI (some are just subsets of AI). While the technology first found its way in the 1940s during the world war through the imitation game of a sole scientist and coined the term 'AI (Artificial Intelligence)' in 1956 at Dartmouth College, Data science found its way in 1974 as the alternative computer science name. Data Science is important because, nowadays, it serves as the foundation for extracting valuable insights, patterns, and predictions from vast amounts of data, driving informed decision-making across various industries, optimizing processes, and enhancing efficiency and effectiveness in business operations. AI and Data science are now integrated and incorporated by many researchers to drive innovation and to improve decision-making.

But that's a vast study of different domains; this review only provides a comprehensive overview of AI technology's diverse applications and implementations across various sectors. Some might get confused with the brief history of AI, but Fig. 7 might clear all the doubt and confusion. The visual representation states the most straightforward flow of AI establishment throughout history.

2.2. Classifications of AI

Based on several factors and perspectives, AI is classified in multiple ways, hence more than one type of AI classification. The classification of AI is not set in stone, and different scholars, researchers, and organizations may have various ways of categorizing AI. Multiple ways exist to put AI systems into computer science and artificial intelligence groups. But putting AI systems into groups based on how smart they are (The level of intelligence) and what they can do (Capabilities of AI systems) is generally accepted and is often used as a starting point to learn about the different kinds of AI systems.

The idea behind this grouping is that AI systems can be grouped based on how well they can do jobs and solve problems. It starts with the broad group "Artificial Intelligence" and then divides into "Weak AI" and "Strong AI." Below is presented a comprehensive classification framework for AI systems based on their levels of intelligence and capabilities, as depicted in Fig. 8. This framework aims to provide a structured understanding of AI systems and their evolving sophistication.

Fig. 8 depicts the classification of AI systems into two overarching categories: Weak AI and Strong AI, with further subdivisions including Artificial Narrow Intelligence (ANI), Artificial General Intelligence (AGI), and Artificial Superintelligence (ASI). In contrast, ANI has moved further with eight more subdivisions.

As illustrated in Fig. 8, "Narrow AI," or "Weak AI," refers to artificial intelligence systems developed specifically to carry out specific tasks inside a particular domain. These artificial intelligence systems have a

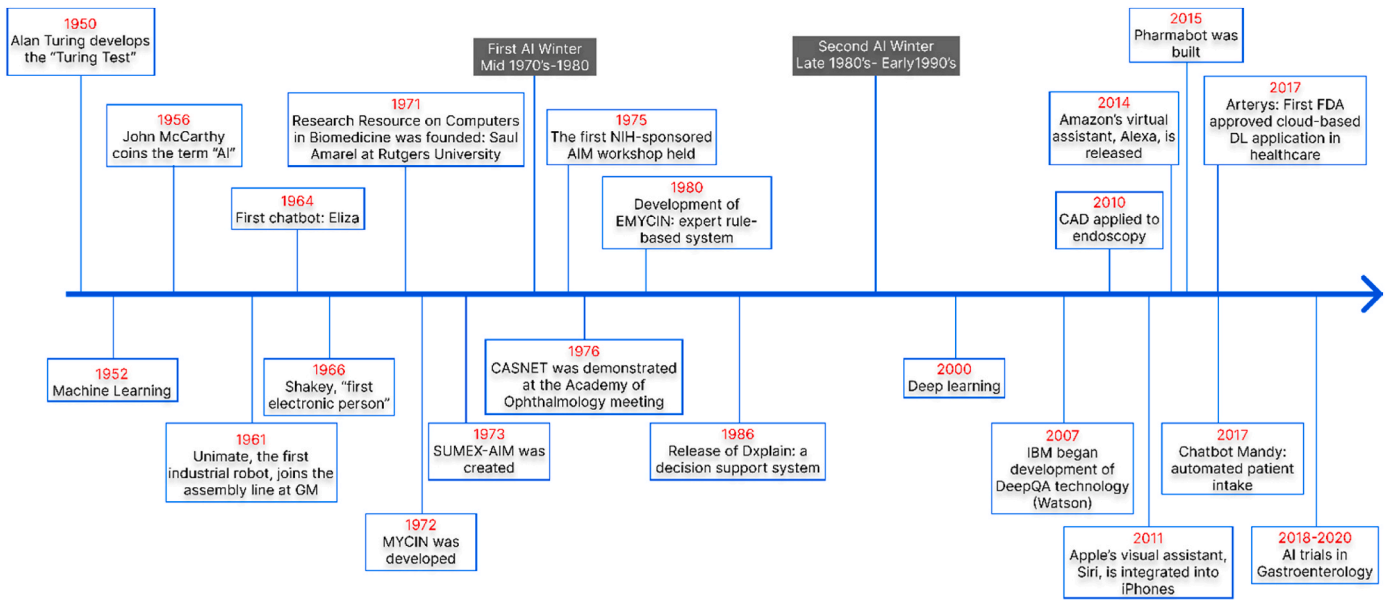


Fig. 7. The continual history of AI [22–28].

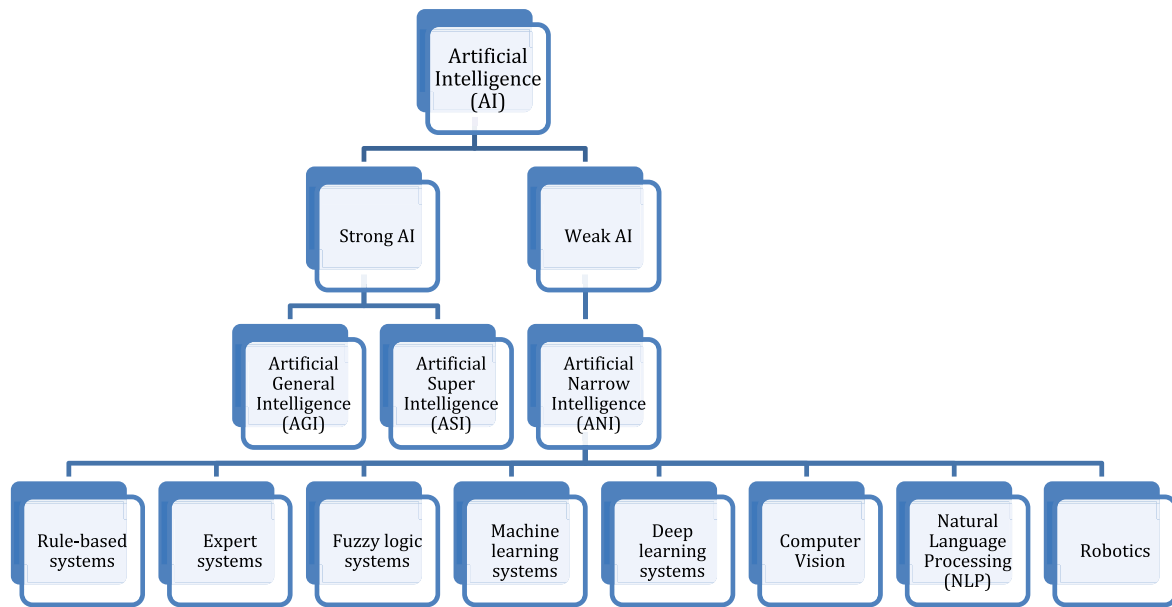


Fig. 8. Classification of AI based on the level of intelligence and capabilities [16].

restricted field of operation and cannot think or reason outside their programming parameters. Examples of weak artificial intelligence include robotics, machine learning, natural language processing, computer vision, and expert systems that rely on rules and predefined answers. Another example is computer vision [29].

Strong AI, on the other hand, refers to an artificial intelligence system capable of reasoning and thinking like a human being and doing any intellectual activity that a human being is capable of. Strong artificial intelligence (AI) includes several subcategories, two of which are artificial general intelligence (AGI) and artificial super intelligence (ASI) [30].

Artificial general intelligence is a system that can perform any task that a person may imagine. A computer program with these capabilities can learn, reason, grasp the natural language, and react appropriately to novel situations. On the other hand, advanced swarm intelligence, or

ASI, is an artificial intelligence system that is superior to humans in every manner.

The classification of artificial intelligence systems based on the amount of intelligence and capabilities they possess is generally acknowledged. This is because it gives a framework for understanding the various types of AI systems and their level of intelligence and capabilities. Academics and industry practitioners alike can zero in on particular facets of artificial intelligence development thanks to the depth and complexity of the occupations in demand.

Furthermore, AI systems are also classified into four major categories based on functionalities, as highlighted in Fig. 9: Reactive machines or robots, Limited memory, Theory of mind, and lastly, Self-awareness, depending on the degree to which they can duplicate human intellect. To mention that The Reactive machine AI systems and Limited AI systems fall under the Weak AI which is also called as Narrow AI and

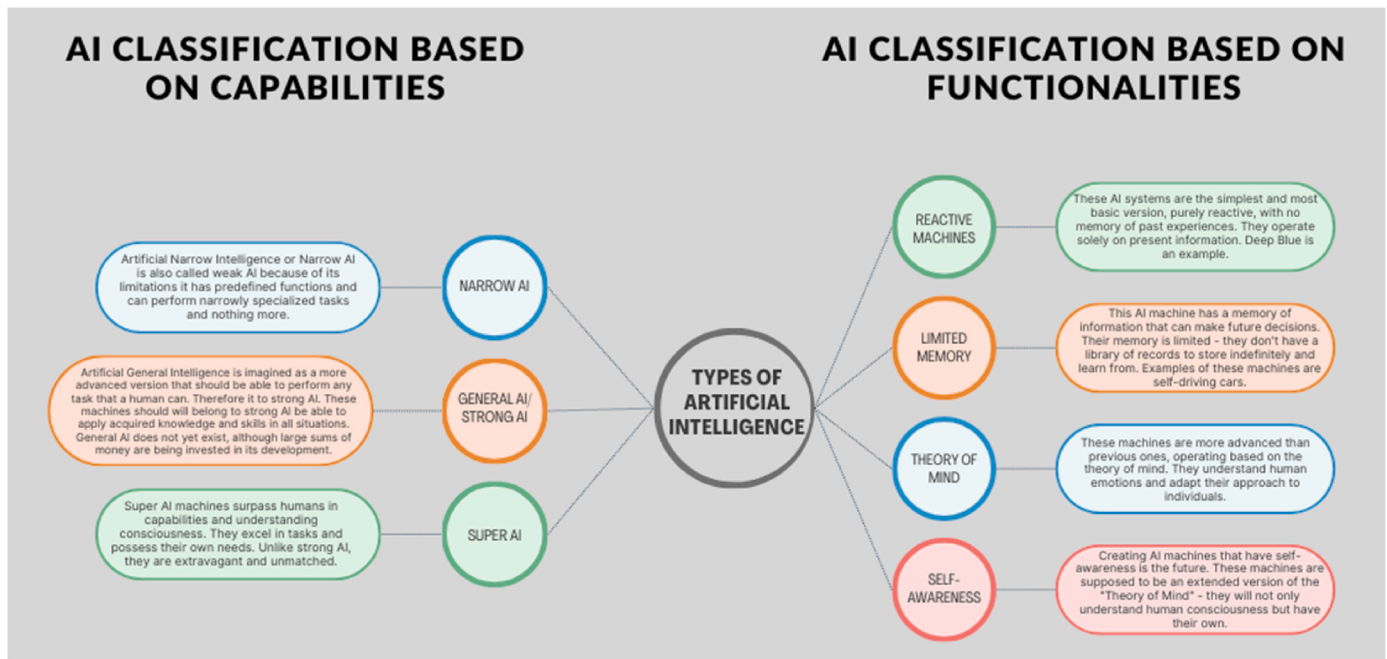


Fig. 9. The classification of AI (Based on application and ways of utilization) [31].

Theory of mind & Self-awareness AI systems (Sentient AI) falls under the category of Strong AI [31].

Reactive machines: In artificial intelligence, reactive machines always produce the same output in response to the same input. This is because reactive robots learn from their past experiences. Most machine learning models are reactive, providing customized suggestions in response to consumer input, such as search or purchase history. In this scenario, the AI behaves reactively. The method relies on "super" artificial intelligence to provide recommendations for users because it is impossible for a human person to go through the whole viewing history of a Netflix subscriber and develop personalized recommendations. Reactive artificial intelligence is reliable and effective in most applications, including goods like self-driving automobiles. The only way it will be able to predict the future is if you provide it with the appropriate information [32].

Limited memory: The next stage in the development of AI is the restricted memory variety. This algorithm mimics how the neurons in our brains communicate, which means it becomes more intelligent as it is given more data to train on [33]. Image recognition is just one of the many types of reinforcement learning that can benefit from deep learning. In contrast to reactive machines, artificial intelligence with limited memory may investigate the past and track certain objects or situations across time. After that, these observations are programmed into the AI so that the AI can perform its activities based on both data from the past and data from the present instant. However, due to the AI's limited memory, this information is not stored within its memory as experience from which it can learn; in the same way, humans can take meaning from their achievements and failures [34]. Artificial intelligence (AI) improves over time as it is taught on more data. Examples of this form of artificial intelligence include autonomous vehicles, unmanned aerial vehicles (drones), and autonomous weapon systems (AWS).

Theory of mind: The first two forms of artificial intelligence mentioned above are currently used. These include reactive machines and limited-memory AI systems. In the future, we can program artificial intelligence with a theory of mind and self-awareness [35]. Because of this, no applications are available. Theorizing about the mind, if artificial intelligence ever becomes a reality, it may one day be able to comprehend what it is like for other beings to have thoughts and feelings

and how the world functions. This, in turn, causes a change in how they interact with other people [36].

Self-awareness: The achievement that would stand out as the most significant in the field of research into artificial intelligence would be the development of systems with a sense of self and a conscious awareness of their existence [37]. This sort of artificial intelligence has not yet been developed. This expands the capabilities of artificial intelligence beyond the comprehension of emotions and into the realms of self-awareness, awareness of one's current state of being, and the ability to recognize or anticipate other people's emotions. The expression "I'm hungry" can be developed into something more particular by adding phrases such as "I know I am hungry" or "I want to eat lasagna because it's my favorite food." We are a long way from creating self-aware artificial intelligence since there is still a great deal to learn about the intelligence of the human brain and how memory, learning, and decision-making work [38].

There are also other types of AI classification apart from these significant two, like rule-based or expert systems AI classification, Connectionist AI classification, Bayesian AI, Hybrid AI, Evolutionary, etc. These classifications are based on several facets of artificial intelligence, including its algorithms, learning processes, and problem-solving techniques. They are frequently utilized in educational settings to inform academics and practitioners about the many AI systems and the potential domain-specific applications that may be accomplished with them.

2.3. Components of AI to function and work

Let's discuss the components of AI. What is inside it, and what type of tools really come into play for AI to function? The truth is, there is no accurate answer to that; however, researchers conclude on these as the basic components of AI:

- Planning.
- Machine Learning (ML).
- Natural Language Processing.
- Computer Vision.
- Robotics.
- Artificial General Intelligence.

By integrating these hypotheses into real-life technologies, artificial intelligence has evolved, reshaping industries and transforming the way we interact with technology.

The process for these components of Fig. 10 to work involves feeding the algorithm with a large dataset, which the algorithm analyzes and learns from, to develop a model that can accurately predict or classify new data. The algorithm automatically identifies patterns and relationships within the data and adjusts the model accordingly to improve its accuracy over time. ML is widely used in various applications such as image recognition, speech recognition, natural language processing, and many more [39].

The components and algorithms such as Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, and Expert Systems are fundamental to the functioning of AI. Machine Learning is a process that involves training a model to make predictions or decisions based on data. The model is trained on a large dataset, and the algorithm automatically learns patterns and relationships within the data. This allows the model to make accurate predictions or decisions on new, unseen data. It is a subset of artificial intelligence that involves training algorithms to learn patterns and relationships within a large dataset to make accurate predictions or decisions on new, unseen data. NLP has numerous applications, including chatbots, language translation software, and virtual assistants such as Siri and Alexa. It has revolutionized how we interact with computers and increased the efficiency and effectiveness of human-machine communication [40].

If we are to discuss the most superficial way how these 8 key components work within the embodiment of AI, Table 1 might clear out everything. Simply 3 steps- Understanding the amalgamation of expert systems, fuzzy logic, neural networks, evolutionary algorithms, and case-based reasoning in hybrid AI systems can be simplified into three steps, facilitating comprehension of their collaborative functionalities.

Expert Systems are AI systems that mimic the decision-making ability of a human expert in a particular domain. These systems use knowledge and rules to reason about problems and make recommendations or decisions. According to Medsker, hybrid intelligent systems [42] are commonly implemented using traditional computing systems, often utilizing microcomputer hardware and software techniques. These systems incorporate various intelligent technologies in conjunction with fuzzy logic when approaching AI.

Constructing intelligent AI systems that can learn, reason, and make decisions requires using various configurations and combinations of the components and algorithms that make up these systems. This includes

Table 1

Intelligent technologies used in hybrid systems of AI while approaching with fuzzy logic [41].

Hybrid Intelligent Systems		
Expert Systems	Fuzzy Logic Evolutionary Algorithm Case-based reasoning	Neural Networks
Hardware & Software		

expert systems, fuzzy logic, neural networks, evolutionary algorithms (such as genetic algorithms, ant colony optimization, and particle swarm optimization), and case-based reasoning. These are inevitable for any AI to function and work properly [43].

3. Literature review

The rapid progression of Artificial Intelligence (AI) technology has catalyzed transformative changes across numerous industries, ushering in an era of unprecedented automation, efficiency, and innovation. Back in 2016, Li et al. [44] did a review study on the application of AI in intelligent manufacturing that highlighted the recent applications of that time and came up with the evaluation of the new era 'Internet plus AI'. Furthermore, that study paper explored the latest developments in intelligent manufacturing technology, industry practices, and practical demonstrations. It provided insights into the ongoing progress in this field and offered recommendations for successfully integrating AI in intelligent manufacturing in China. While the former study failed to give an overview of other sectors (studied only the manufacturing sector), a recent systematic study of 2022 discussed the ongoing revolution of AI technologies and their impact on various sectors [45]. However, their study was more focused on existing challenges and required principles or policies to uplift the innovation of AI to a new height. Another review study from the same year offered a thorough examination of AI and big data within the context of Industry 4.0 [46]. It specifically focused on key applications, techniques, underlying concepts, enabling technologies, challenges, and research perspectives related to the transition to Industry 5.0. The study delved into how combining AI and big data contributes to various applications within Industry 4.0. However, their study was limited to theoretical discussions of policies to be integrated and conceptual frameworks without an overview of real-world case studies of those industries of different organizations.

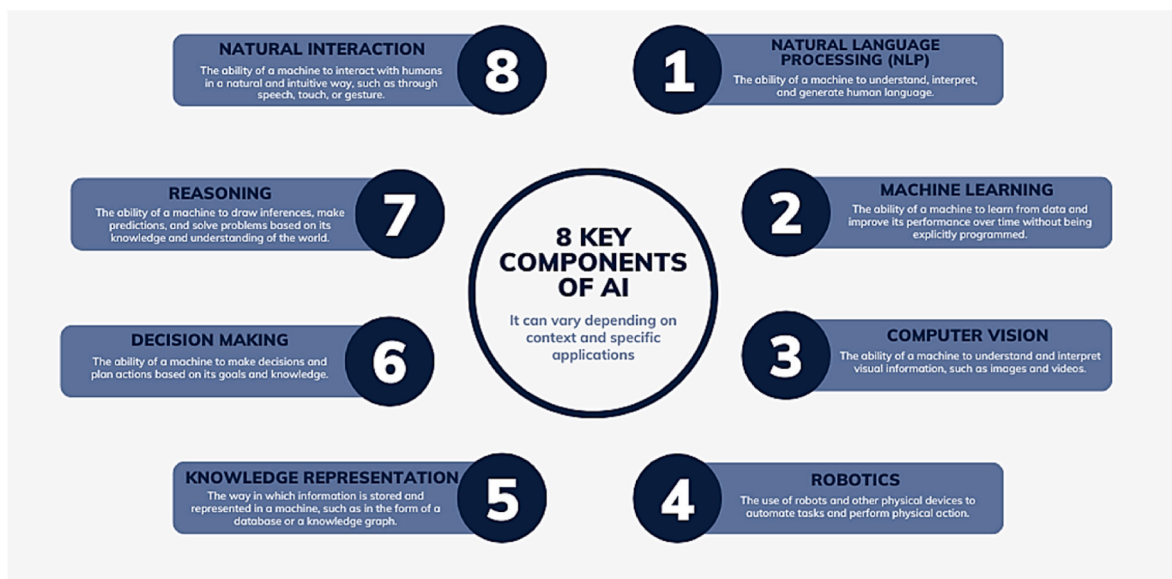


Fig. 10. 8 Key components of AI which are variable depending on a specific application.

A study by Iqbal H. Sarker offers a comprehensive overview of AI-based modeling in Industry 4.0 [47]. It highlights various types of AI and their applications across industries. The paper aims to serve as a reference guide for academics, industry professionals, and decision-makers, providing insights into developing intelligent systems to address real-world challenges. However, it also acknowledges the inherent challenges in developing effective AI models due to the dynamic nature of real-world problems and data variability. Also, the study focuses more on AI-based modeling than applications and implementation overviews [47]. Anantrasirichai and Bull's study reviews AI applications in creative industries, discussing techniques like CNNs and GANs in 2022 [48]. It categorizes AI use into content creation, analysis, enhancement, extraction, and data compression. It notes AI's role as a tool for human creativity but highlights challenges in achieving AI-generated creativity comparable to humans, emphasizing the need for human-centric AI design. However, the study underscores the growing importance of AI in creative industries, suggesting a shift towards collaborative human-AI creativity rather than AI as a standalone creator.

The literature-based study by Javaid et al. [49] focuses on the vital role of Artificial Intelligence (AI) in Industry 4.0, focusing on its contributions to improving productivity, product consistency, and reducing operating costs. It highlights the collaborative partnership between robotics and humans, emphasizing the need for AI automation systems to interpret various data types and facilitate decision-making in smart industries. The paper discusses the technological features, advancements, challenges, and applications of AI for Industry 4.0, showcasing its wide-ranging benefits, such as predictive maintenance, quality management, and optimization of manufacturing processes. Additionally, it underscores the significance of AI in revolutionizing production through its integration with other major Industry 4.0 technologies, ultimately driving towards more efficient and data-driven manufacturing practices. Overall, the study is not too deep in mentioning the applications and implementations of AI. The study was more concerned with analyzing how AI drives Industry 4.0 [49].

A backdated study in 2020 [50] greatly impacted the showcase of AI's applications and implementations, but that was also focused on industrial AI's manufacturing domain and design principles. The latest study on applications and implementations of AI could be mentioned as the study of [51], which addressed research questions like: What are the concerns that current Industrial Artificial Intelligence (IAI) techniques address in Industry 4.0 (I4.0)? Which AI techniques are used to address which concerns? What are the challenges to the adoption of current IAI technologies in I4.0? However, it is limited by the scope of the literature review, which may not encompass all relevant studies, particularly those published in non-academic sources such as industry reports or trade journals. The authors highlight the significance of AI and machine learning technologies in driving Industry 4.0 initiatives, leading to the emergence of Industrial Artificial Intelligence (IAI). They discuss how AI technologies have revolutionized various industry sectors, such as manufacturing, finance, transportation, healthcare, and science, by enabling predictive maintenance, supply chain optimization, and digital twins [51]. In the perspective of only three AI technologies.

3.1. Evolution of industry 4.0 and emergence of industry 5.0 focusing AI

The research case of [52] gave a concise idea of the situation of Industry 4.0 and Industry 5.0, mentioning AI as a major enabling technology; however, it failed to dive deeper. The industrial landscape has profoundly evolved with the advent of Industry 4.0, characterized by the convergence of AI, Big Data, the Internet of Things (IoT), and Cyber-Physical Systems (CPS). Industry 4.0 aims to optimize manufacturing processes, enhance productivity, and drive operational efficiency through advanced technologies [53,54]. The review [55] examines using artificial intelligence techniques, including supervised and unsupervised learning, among others, in wearable devices.

However, the transition to Industry 5.0 marks a paradigm shift towards human-centric collaboration between humans and AI-powered systems [56]. This shift emphasizes the integration of AI technologies to augment human capabilities while prioritizing societal value and sustainability [57]. Industrial manufacturers invest more in perceptive, predictive, and prescriptive AI capacity development for enhanced applications [58].

3.2. Foundations and technologies of industry 4.0 and 5.0

2021 was when the technological competitiveness and emerging technologies in both Industry 4.0 and Industry 5.0, like AI were elaborately highlighted [59]. A suite of cutting-edge technologies, including AI, Big Data analytics, IoT sensors, and Blockchain [60], underpins Industry 4.0. These technologies enable predictive maintenance, smart manufacturing, and real-time data analytics to drive informed decision-making [2]. The emergence of Industry 5.0 builds upon these foundations, fostering human-machine collaboration and emphasizing a holistic approach to value creation [11].

3.3. AI in industry transformation

A case study of 2022 (Akter et al., 2022) analyzed the application of AI, blockchain, and cloud & data analytics to evaluate how digital innovations transform businesses. AI is a cornerstone of Industry 4.0 and 5.0, driving digital transformation across diverse sectors. Industries can automate processes, optimize resource utilization, and enhance decision-making by harnessing AI capabilities such as machine learning, deep learning, and natural language processing [61]. AI-powered solutions have revolutionized areas such as healthcare diagnostics, financial forecasting, and manufacturing operations, unlocking new levels of efficiency and productivity [62].

3.4. Current state and Future Directions of AI

Undoubtedly, AI embodies the forefront of industrial digitalization and digital servitization, marking the next evolutionary leap in technological advancement [58]. 1983 was the first time that a highlight was made to discuss the problems that confront AI and answer some questions about the field [63]. The latest state was not summarized in any research case or reviews rather, it highlighted only specific domains like businesses [64], education [65], life science [66], image processing [67], healthcare [68], autonomous driving [69], monitoring [70], robotics [71], manufacturing [72], and others. AI technologies permeate various facets of daily life and industry operations. Machine learning algorithms power recommendation systems, autonomous vehicles, and predictive analytics platforms, driving efficiencies and enhancing user experiences [21]. Looking ahead, the future of AI holds promises of continued innovation, with advancements in deep learning, neural networks, and natural language processing poised to unlock new frontiers in AI-driven applications (Shrestha & Mahmood, 2019).

Fig. 11 here visualizes the databases and registers that were gone through during the review study. In summary, the literature review highlights almost all the recent and previous most crucial review studies and case studies that highlighted and summarized the applications and implementations of AI, which played a pivotal role in driving industrial transformation from the advent of Industry 4.0 to the human-centric ethos of Industry 5.0.

It is found that most studies are focused on specific domains of AI applications or more than one. But do not classify the applications of AI to enhance the study and do not highlight that 'AI is everywhere now'. By synthesizing historical perspectives, technological advancements, and current trends, this review sets the stage for a comprehensive understanding of AI's multifaceted impact on industries worldwide in almost all sectors.

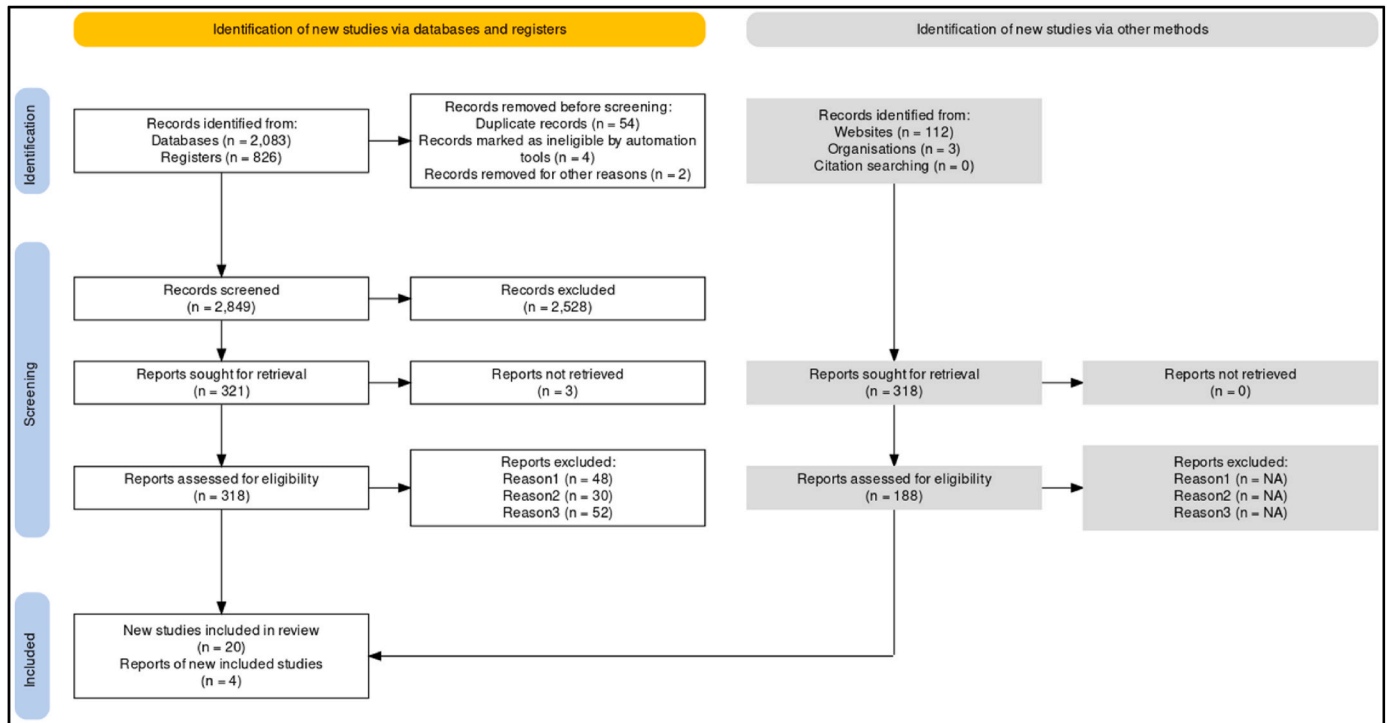


Fig. 11. Prisma Flow diagram of the review based on databased and registers.

4. Applications and implementations of AI

Artificial Intelligence (AI) has become a critically important technology in recent years, with a broad range of applications across multiple industries. It has become integral to numerous fields, revolutionizing industries and addressing pressing global challenges. Here's a glimpse into some of the key areas leveraging AI advancements per classification:

- **AI for Earth and Environmental Conservation:** Employing AI to tackle sustainability issues such as resource management, biodiversity preservation, and combating climate change.
- **AI for Social Impact:** Harnessing AI's capabilities to address complex social issues, including cancer diagnosis, identifying online exploitation victims, and aiding in disaster response efforts.
- **AI in Music:** Transforming the music industry by automating tasks, analyzing vast datasets, and enhancing creative processes.
- **AI in Art:** Enabling computational creativity through AI-driven tools for tasks like discovering mathematical principles, generating poetry, and producing visual art.
- **AI in Sports:** Enhancing sports performance and management across various disciplines, from real-time analytics to athlete training and advertising strategies.
- **AI Ethics:** Guiding the development of AI systems with ethical principles to ensure responsible decision-making and behavior.

As AI technology continues to evolve, its impact on diverse sectors is poised to expand exponentially. What was once confined to science fiction is now a tangible force shaping the future. The potential for AI to transform our world is boundless, with new applications emerging rapidly, promising to revolutionize industries and improve lives worldwide.

Let's have an overview of that part, the overall applications and implementations of AI globally.

4.1. AI in the agriculture sector

AI in the global agriculture market is forecasted to grow at a CAGR of approximately 25 % from 2023 to 2031. AI applications in precision farming have boosted crop yields by 20–30 %. Additionally, AI in irrigation systems has cut water usage by up to 25 %. Similarly, AI-driven fertilizer and pesticide application solutions have improved efficiency and reduced input usage [73].

Humans have reached the limit of our cognitive abilities due to the development of technology, and we are now attempting to combine our natural brain with an artificial one. This ongoing investigation gives rise to a brand-new industry called artificial intelligence in agriculture. The countries and nations were primarily involved and focused on the issue of agriculture and the development of the agricultural sector through automation. There is an urgent need for food, given how quickly the global population is growing (7.9 billion by the end of 2022), and it is projected to climb more in the following days. Because their current methods cannot keep up with the expanding demand, farmers are forced to employ toxic pesticides and other chemicals to improve productivity and, more significantly, to implement them more frequently. This ultimately damages and harms the soil; over time, it costs the agricultural industries much money. The land continues to be unproductive and barren because of the significant adverse effects of farming practices [74]. AI has come in assistance to solve these types of issues.

4.1.1. AI in agricultural management

AI has been applied in agriculture like soil management, crop management, testing, hybrid seeds production, new variants of crops for better production, apps to sell the crops in different markets, networking options for better pricing, online servicing, efficient irrigation process, greenhouse management, and many others.

Table 2 showcases various AI-driven solutions revolutionizing farming practices. From precision irrigation to weed detection and crop health monitoring, these technologies optimize resource usage and enhance productivity for sustainable agriculture.

Table 2
Diverse applications of AI-based technology in advancing and developing the agricultural sector worldwide [75].

Ser No.	Technology	Applications/Implementations	Sources
1	CropX	This AI-powered platform uses soil sensors and satellite imagery to optimize irrigation and fertilization, improving crop yields and water conservation.	[76]
2	FarmWise	FarmWise develops autonomous robots that are modelled and run by computer vision and ML algorithms to detect and remove weeds, reducing chemical usage and increasing crop yield.	[77]
3	Blue River Technology (acquired by John Deere)	Blue River's See & Spray technology employs computer vision and AI to identify and precisely apply herbicides only to weeds, minimizing chemical usage and promoting sustainable farming practices.	[78]
4	Taranis	Taranis utilizes AI and high-resolution imagery captured by drones to monitor crop health, detect pests and diseases, and provide farmers with actionable insights for optimized crop management.	[79]
5	Prospera	Prospera's AI-powered platform combines cameras, sensors, and machine learning algorithms to monitor crop health, detect anomalies, and optimize irrigation and fertilization strategies for increased yields and resource efficiency.	[80]
6	Gamaya	Gamaya's hyperspectral imaging technology and AI algorithms provide farmers with detailed insights into crop health, nutrient deficiencies, and pest infestations, enabling targeted interventions and improved crop management.	[81]
7	AgriWebb	AgriWebb offers a farm management software solution that leverages AI and data analytics to optimize livestock management, track animal health, and improve farm productivity.	[82]
8	Trace Genomics	Trace Genomics uses AI and genomic analysis to assess soil health, identify beneficial microbes, and recommend personalized soil management practices to optimize crop growth and resilience.	[83]
9	Climate Corporation (acquired by Bayer)	Climate Corporation's Climate FieldView platform utilizes AI and data analytics to provide farmers with field-specific weather forecasts, crop modeling insights, and agronomic recommendations for optimal decision-making.	[84,85]
10	Arable Labs	Arable's IoT-enabled crop monitoring devices, combined with AI algorithms, deliver real-time data on weather conditions, soil moisture levels, and crop health metrics to help farmers make informed decisions and maximize yields.	[86,87]

4.1.2. The use of AI-augmented drone in agriculture

Fig. 12 illustrates the four-step procedure of disease detection of crops with the help of AI. Overall, throughout the whole world, AI has brought a revolution in agriculture. According to a recent study by PWC (Price Water House Coopers), the global market for drone-based solutions is worth \$127.3 billion [88]. In addition, the amount allocated to

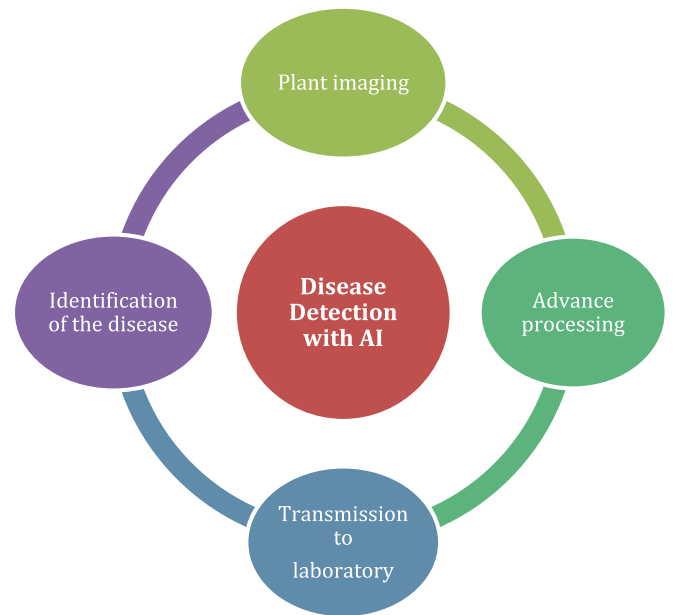


Fig. 12. Disease detection process with AI [89].

agriculture is \$32.4 billion [75].

Such drone-based solutions in the agricultural sector have the potential to assist with a wide variety of issues, including mitigating the effects of unfavorable weather, raising production levels, farming with greater accuracy, and optimizing crop yields [89]. Aside from the significant use of AI-augmented drones in agriculture, currently, robotics (one of the AI analytics) is also being used in digital farming:

Precision farming by robotics (Fig. 13), an AI-developed farming method, eliminates the need for the tedious and manual labor associated with conventional farming. Precision farming is a more precise farming technique that provides valuable advice on crop rotation. Technologies such as high-precision positioning systems, geological mapping, remote sensing, integrated electronic communication, variable-rate technology, an optimum planting and harvesting time estimator, water resource management, plant and soil nutrient management, and pest and rodent attack management [90] are all integral parts of precision farming [91].

4.1.3. Agricultural tool enahncement with AI

Tractors are another inevitable necessity in the farming industry. There is also some rapid transformation with the help of AI. Here are some specific AI-based tractors, as illustrated in Fig. 14 that are already in the market and used in agriculture:

- **Monarch Tractor's MK-V:** This driver-optional, all-electric tractor utilizes deep learning capabilities for real-time data processing, adjustments, and long-term yield estimates [95].
- **John Deere's Autonomous Electric Tractor:** Set to launch in Australia in 2026, this autonomous electric tractor will feature local trials beforehand. It's expected to revolutionize Australian agriculture with its autonomous capabilities [96].
- **Yanmar's e-X1 Electric Tractor:** Yanmar's electric tractor concept, the e-X1, can be controlled remotely and is designed for use in extreme or hazardous environments. It may also incorporate autonomous driving features in the future [97].

The use of AI in agricultural problems needs to become more commercially viable before the farming community can benefit from this technology. If the AI cognitive solutions were made available on an open-source platform, this would lower the price of the solutions, eventually leading to a higher rate of adoption and improved insight

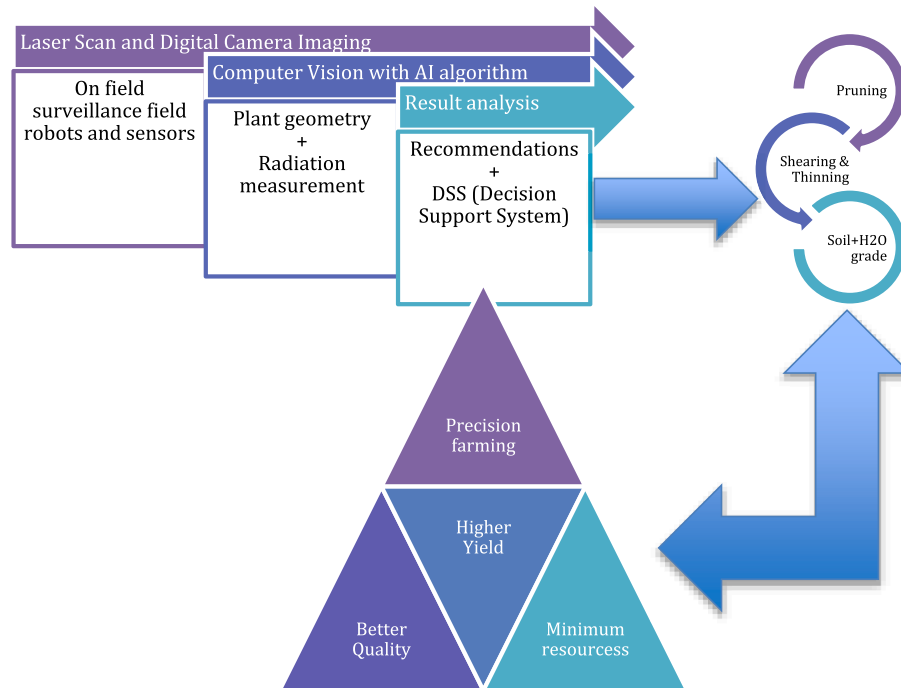


Fig. 13. Robotics process application in digital farming [89].



Fig. 14. Latest AI-augmented autonomous tractors worldwide that are helping the agriculture sector to advance, (a) MK-V [92], (b) JD9X [93], (c) Yanmar's e-X1 [94].

among farmers.

4.2. AI in the education sector

Artificial intelligence, personal computers, and other assistive technologies can be integrated into robots to enable the development of robots that improve student learning, starting with the most fundamental type of schooling, teacher instruction in preschool.

It is suggested that cobots, or the application of robots, working alongside teachers or colleague robots, are being used to teach youngsters everyday activities, like spelling and pronunciation, and modifying to the student's skills. Similarly, the delivery of materials online or locally so that students may transfer, study, and finish their assignments to pass has improved and now includes intelligent and adaptable web frameworks. Determine the behavior of the teacher and the students and make necessary changes to improve the educational experience.

4.2.1. The AI-based techs in the education sector

According to Chassignol, school administration, instruction or teaching, and learning have all been combined with artificial intelligence [98].

- **Chatbots for Student Support:** AI-powered chatbots instantly respond to student inquiries, offering guidance on course registration, assignment deadlines, and academic resources [99].

- **Automated Essay Scoring:** AI algorithms assess and score student essays based on factors such as grammar, structure, and content coherence, providing rapid feedback to both students and educators [100].
- **Speech Recognition Software:** AI-driven speech recognition software transcribes spoken language into text, facilitating language learning, speech therapy, and accessibility for students with hearing impairments [101].
- **Virtual Classroom Simulations:** AI-generated virtual classroom simulations replicate real-world teaching scenarios, allowing educators to practice instructional techniques, classroom management strategies, and student engagement methods [102].
- **Predictive Analytics for Student Success:** AI-powered predictive analytics models analyze historical student data to forecast academic outcomes, identify at-risk students, and recommend interventions to improve retention and graduation rates [103].
- **Adaptive Content Delivery Systems:** AI algorithms dynamically adjust the difficulty and pace of learning materials based on student performance and comprehension levels, ensuring optimal learning progress for each learner [104].
- **Emotion Recognition Technology:** AI-driven emotion recognition technology detects and analyzes facial expressions and body language, providing insights into student engagement, motivation, and emotional well-being during online learning sessions [105].

- **Intelligent Content Creation Tools:** AI-based content creation tools generate educational materials, such as quizzes, worksheets, and lesson plans, tailored to specific curriculum requirements and learning objectives [106].
- **Personalized Learning Pathways:** AI-powered personalized learning platforms curate customized learning pathways for students, recommending relevant courses, modules, and learning resources aligned with their academic goals and interests [107].
- **Simulated Learning Environments:** AI-driven learning environments simulate real-world scenarios, such as medical simulations for healthcare education or virtual laboratories for science experiments, allowing students to practice skills and apply knowledge in risk-free settings [102].

Here are some more recent and specific techs that AI drives to help the education sector grow:

Table 3 illustrates AI's role in education, showcasing various platforms like Jill Watson and Squirrel AI Learning, offering personalized learning paths, tutoring, and adaptive instruction for students worldwide.

Table 3
Latest AI-augmented tech products revolutionizing the Education sector worldwide [108].

SI No.	Technology	Applications/Implementations	Sources
1	Jill Watson	An AI-enabled virtual teaching assistant introduced by the Georgia Institute of Technology in 2016	[109]
2	Squirrel AI Learning	An adaptive learning platform that utilizes AI algorithms to personalize learning paths for students and provide tailored feedback and support.	[110]
3	Knewton	An adaptive learning engine that analyzes student data to recommend personalized learning materials and adaptively adjust content difficulty levels.	[111]
4	Brainly	Social media platform for questions	[112]
5	Thinkster Math	It is a math tutoring program that combines human interaction with AI to develop personalized learning plans.	[113]
6	Cognii	An AI-powered virtual learning assistant designed to engage students in natural language dialogue, provide personalized tutoring, and offer real-time feedback on open-ended questions.	[114]
7	Querium	An AI-powered tutoring platform that delivers personalized instruction and assessment for students in math and science subjects.	[115]
8	Edmodo	A social learning network that integrates AI features to facilitate collaboration, communication, and personalized learning experiences for students and teachers.	[116]
9	Acellus Academy	An online learning platform incorporating AI to deliver interactive course content, assessments, and personalized learning paths for K-12 students.	[117]
10	DreamBox Learning	An adaptive math learning platform that uses AI algorithms to provide students with individualized instruction and adaptive practice.	[118]
11	TutorMate	An AI-powered literacy intervention program that pairs students with volunteer tutors for remote reading sessions, leveraging AI for personalized reading support.	[119]
12	Carnegie Learning	A provider of AI-driven math curriculum and instructional software that adapts to individual student needs and provides real-time feedback to teachers.	[120]
13	IXL Learning	An adaptive learning platform that uses AI to provide personalized practice in math, language arts, science, and social studies for K-12 students.	[121]

Fig. 15 shows an AI model that uses machine learning and cognitive modeling to provide students with individualized and dynamic instruction. The tutor's cognitive model for each student allows for constant revision and adaptation of lessons in response to the student's actions and progress. This way, the adaptive tutor can determine which areas need more attention, modify the difficulty level accordingly, and provide recommendations, comments, and learning materials unique to each student. By evaluating student data and using algorithms like knowledge tracing and adaptive feedback generation, AI improves the classroom experience by providing students with engaging, personalized lessons [123].

4.2.2. Intelligent tutoring systems

Intelligent tutoring systems like Carnegie Learning's Mika platform are a great example of the use of AI in this sector [124]. This platform combines artificial intelligence (AI) and cognitive science technology to offer real-time feedback and individualized tutoring to post-secondary students, particularly new first-year college students who might otherwise require remedial courses. According to Carnegie, remedial education costs universities \$6.7 billion yearly, and only 33 % of students succeed in math courses. With ITS, students may have easier, continuous access to flexible and more tailored learning methods.

Every year, there is more and more interest in using AI algorithms and systems in education around the globe; even third-world countries are also trying to adopt AI in their educational system. Since 2010, there has been a sharp increase in the number of papers published on "AI" and "Education" on the Web of Science and Google Scholar.

The papers published between 2015 and 2019 comprised roughly 70 % of all the publications indexed. Scientists and many prolific researchers work to use the most advanced AI model approaches, such as deep learning and data mining, dealing with complex problems and customizing teaching methods for each student, as these technological advances in education have a more significant impact on how well people learn. Because the nature of higher education is changing rapidly, more than half of institutions and colleges now rely on ML (a subdomain of artificial intelligence) for authoritative assistance [21].

4.2.3. Specification on the AI contribution

To sum up, AI in the education sector has helped evolve in:

- Personalized learning.
- Smart content creation.
- Task automation.
- Digital lesson generation.
- Adaptable access to AI software or ML-based classes.
- Determining classroom vulnerabilities.
- Closing skill gap.
- 24/7 assistance and guidance by conversational AI.
- Securing and decentralizing learning systems.
- AI in examinations.

There are more applications of AI in this part, and day by day, it's increasing more. Many countries are even considering remote education with the help of AI. These AI-assisted education development solutions are affordable, incredibly versatile, and adaptable to the newest instructional approaches.

4.3. AI in the manufacturing sector

There are several difficulties in designing engineering and manufacturing processes and systems, including complexity, unpredictable behavior, and dynamism. Big data, rapid processing, cloud computing, and artificial intelligence methods (including machine learning and deep learning) have recently changed how many engineers and industrial experts approach their work. These inventions provide professionals and manufacturers with stimulating and forward-thinking

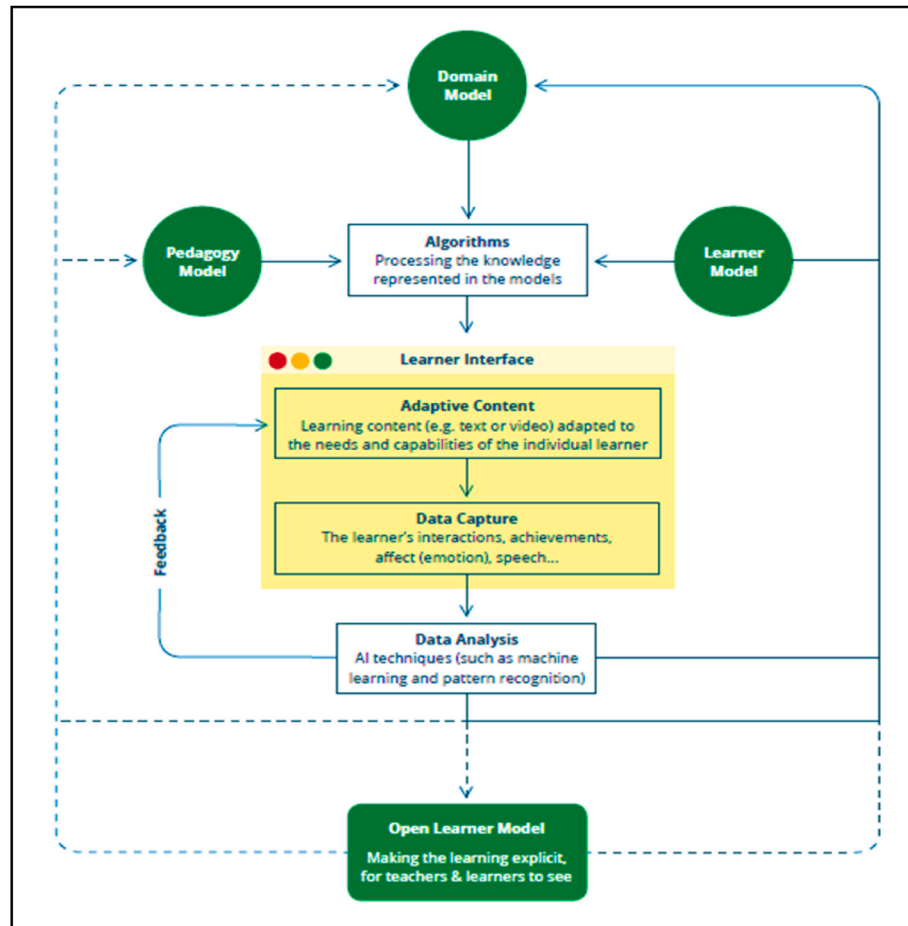


Fig. 15. AIEd system showing a simplified picture of a typical model-based adaptive tutor [122].

solutions to challenging real-world problems.

Choosing the finest artificial intelligence strategy for the correct plan or assembly process can be challenging due to the abundance of available hypotheses, computations, and approaches. Nonetheless, whether we pass this test, artificial intelligence is already overpowering the assembly scene and contributing in several ways [125].

4.3.1. Use of AI in manufacturing process and asset optimization

AI is extensively used in manufacturing for process optimization and asset optimization, which involves several critical applications:

1. **Function Approximation and Model Building:** AI is used to create models for objective and constraint functions. Simple feedforward Artificial Neural Networks (ANN) are often sufficient for these purposes, enabling manufacturers to approximate functions and build accurate models for various processes.
2. **Design Optimization:** AI-based global optimization methods, such as Genetic Algorithms (GA), Evolutionary Algorithms (EA), Particle Swarm Optimization (PSO), and Ant Colony Optimization (ACO), is employed to handle mixed-integer variables, discrete variables, and nonlinearities. These methods ensure convergence to a global optimum. In manufacturing, multicriteria-constrained optimization is commonly performed, targeting objectives such as cost minimization, strength and stiffness maximization, and path optimization.

These AI applications enhance the efficiency and effectiveness of manufacturing processes, contributing to significant advancements in the industry. The incorporation of AI technologies facilitates the development of more precise, efficient, and adaptable manufacturing

systems, ultimately revolutionizing the sector.

Manufacturers can identify situations that could lead to breaking and act before they occur following these six steps, as mentioned in Fig. 16, in any manufacturing domain, including welding manufacturing, shipyard manufacturing, automotive manufacturing, pharmaceutical, textile, FMCG or any other with the help of AI. Manufacturers can forecast the useable life of equipment using machine learning algorithms and prepare it for additional repairs. Robotic Process Automation (RPA) is an AI-based software (mainly integrated by ML) that undertakes repetitive, high-volume processes that traditionally require human staff. These duties could involve keeping track of records, responding to inquiries, performing computations, and more. Modern AI-based bots are introduced to handle industries' labour-intensive and time-consuming tasks, freeing operators for more valuable and sophisticated industrial tasks [126]. Fig. 12 shows an AI model for manufacturing automation with all the included steps.

4.3.2. The field of AI applications in manufacturing world

The field of intelligent manufacturing benefits from AI by developing new models, methods, and forms, as well as current system architecture and technological systems, which are more optimized and efficient for time, money, and labor.

These are just the scratch-bottom of AI in manufacturing, as shown in Table 4. It has helped to manufacture in more new ways in the 21st century than anyone could imagine. In conclusion, the integration of AI technologies in manufacturing, exemplified by predictive maintenance systems and robotic process automation, revolutionizes production processes, optimizing efficiency and resource utilization across various industries, and heralding a new era of innovation.



Fig. 16. Robotic Process automation common steps in manufacturing (Image courtesy: Predictive analytics today).

Table 4
Latest AI-based technologies that are renovating the manufacturing industry.

SI No.	Technology	Application/ Implementation	Company/ Organization	Source
1	ABB Ability™ Connected Services	Predictive maintenance and process optimization	BMW, Volkswagen, General Motors	[127]
2	Siemens MindSphere	Predictive maintenance, production optimization	Ford, Airbus, General Electric	[128]
3	Predictronics	Predictive maintenance solutions	Boeing, Honda, Procter & Gamble	[129]
4	C3.ai Predictive Maintenance	Predictive maintenance solutions	Shell, 3 M, Enel	[130]
5	General Electric (GE) Predix	Predictive maintenance, asset optimization	Caterpillar, Boeing, ExxonMobil	[131]
6	IBM Watson IoT	Predictive maintenance, quality management	Samsung, Airbus, Toyota	[132]
7	Microsoft Azure IoT	Predictive maintenance, remote monitoring	Rolls-Royce, Honeywell, Schneider Electric	[133]
8	PTC ThingWorx	Predictive maintenance, asset optimization	Bosch, Caterpillar, Johnson Controls	[134, 135]
9	SAP Predictive Analytics	Predictive analytics for forecasting, anomaly detection	BMW, Nestle, Coca-Cola	[136]
10	Rockwell Automation FactoryTalk Analytics	Industrial automation and manufacturing processes	Procter & Gamble, PepsiCo, Kimberly-Clark	[137]

4.4. AI in the healthcare and medical sector

Contemporary healthcare institutions commonly employ AI-enabled improvements to enhance the quality and adequacy of clinical resources [274]. Information-escalated medical care provides excellent prospects for development because of AI-based innovations that integrate AI, natural language processing, and sophisticated mechanical technology. At the RSNA (Radiological Society of North America) meeting in Chicago in December 2018, numerous new companies and well-designed imaging gadget makers displayed their AI-modelled drives. These drives provide accurate and trustworthy patient assessments and appropriate patient care based on data acquired from clinical evaluations [138].

4.4.1. The initiation of AI applications in the field of healthcare

During the 1970s, interest in biomedical applications of AI exploded [139]. Since then, the journey of AI in healthcare and medicine has never been less. The best examples are Buoy Health, Babylon, PathAI, Tempus, and Spring Health. Each year, around 400,000 hospitalized patients suffer preventable harm, with 100,000 fatalities. AI has significant potential to improve diagnoses, reducing fatal human errors caused by heavy caseloads and incomplete medical histories. Unlike humans, AI can quickly and accurately identify and predict diseases. The general idea behind these medical innovations is to employ PC calculations to filter pertinent data from information and assist clinical navigation, regardless of the precise artificial intelligence methods. AI advancements are also suitable for other healthcare jobs in the field including assisting with age and treatment selection, determining a disease's risk and severity, reducing therapeutic error rates, and enhancing effectiveness [140].

Demographic information, notes from medical professionals, pictures, test results from labs, information from genetic tests, and recordings from wearable sensors or medical equipment can all be considered personal health data. This data may be generated or collected using various technological platforms, including network servers, electronic health records (EHRs), genomic data, personal computers, smartphones, mobile apps, wearable technology, and sensors. Data dissemination and access have also become simpler with the blessing of technologies, such as cloud-based capabilities and enhanced worldwide communication via the Internet. From integrating AI-augmented big data related to health and disease, unprecedented opportunities exist. Let's look at the latest tech products revolutionizing the healthcare industry through AI for brief applicability and implementation.

Table 5 provides an overview of various AI-based technologies used in healthcare and the medical sector, their applications, the companies or organizations utilizing them, and the respective sources.

4.4.2. Healthcare organizational implementation of AI

AI plays a vital role in several ways in various governmental, non-governmental, and privately run therapeutic environments worldwide. For instance, a triage feature is included in one of the known apps called Babylon, an AI-based chatbot now being tested in the UK. It is used to distinguish between patients of the clinic or hospital who only want reassurance and, on the other hand, those who need a referral for an in-person evaluation. The healthcare system would benefit from AI-based triaging if it could focus resources on patients most likely to need them [140].

Since the 1970s, AI in healthcare has aimed to improve diagnosis and treatment, starting with Stanford's MYCIN [160] for bacterial infections. Early systems like MYCIN showed promise but weren't adopted due to limited advantages over human diagnosticians and poor integration with clinical workflows. Recently, IBM's Watson has gained attention for its focus on precision medicine, especially in cancer detection and treatment [161]. Some AI-augmented detection models, as shown in Fig. 17, make the precision more authentic and less time-consuming.

Table 5
Applications of AI in Medical and Healthcare (mentionable techs) [141].

SI No.	Technology	Application/ Implementation	Company/ Organization	Source
1	IBM Watson Health	Medical diagnosis, treatment recommendations, drug discovery	Mayo Clinic, Memorial Sloan Kettering Cancer Center, Cleveland Clinic	[142]
2	Google DeepMind Health	Medical imaging analysis, patient monitoring	Moorfields Eye Hospital NHS Foundation Trust, University College London Hospitals NHS Foundation Trust	[143]
3	NVIDIA Clara	Medical imaging analysis, drug discovery	National Institutes of Health, Massachusetts General Hospital	[144]
4	Microsoft Healthcare Bot	Patient engagement, virtual health assistants	Providence St. Joseph Health, Novant Health, Premiera Blue Cross	[145]
5	Tempus	Precision medicine, genomic sequencing	Mayo Clinic, Cleveland Clinic, University of Chicago Medicine	[146]
6	Zebra Medical Vision	Medical imaging analysis, early disease detection	Stanford University Medical Center, Oxford University Hospitals NHS Foundation Trust	[147, 148]
7	PathAI	Pathology diagnostics, cancer detection	Beth Israel Deaconess Medical Center, Cleveland Clinic, Bristol Myers Squibb	[149]
8	Arterys	Medical imaging analysis, cardiovascular disease diagnosis	UCSF Medical Center, Stanford Health Care, Ohio State University Wexner Medical Center	[150]
9	Butterfly Network	Point-of-care ultrasound imaging, portable medical devices	Cleveland Clinic, Yale New Haven Health, Northwestern Medicine	[151]
10	Verily Life Sciences	Health data analytics, disease prevention	Duke University Health System, Mayo Clinic, Sanofi	[152]
11	Cerner HealtheIntent	Population health management, predictive analytics	Advocate Aurora Health, Children's Mercy Kansas City, Duke University Health System	[153]
12	Caption Health	Ultrasound imaging, cardiac diagnostics	Northwestern Medicine, Houston Methodist, University of California, San Francisco	[154]
13	Qventus	Hospital operations optimization, patient flow management	Stanford Health Care, Mercy Health, UC San Diego Health	[155, 156]
14	Viz.ai	Stroke diagnosis, neuroimaging analysis	Mount Sinai Health System, OhioHealth, Ohio State University Wexner Medical Center	[157]
15	Aidoc	Medical imaging triage, prioritization	Yale New Haven Health, University of Rochester Medical Center, Sheba Medical Center	[158]

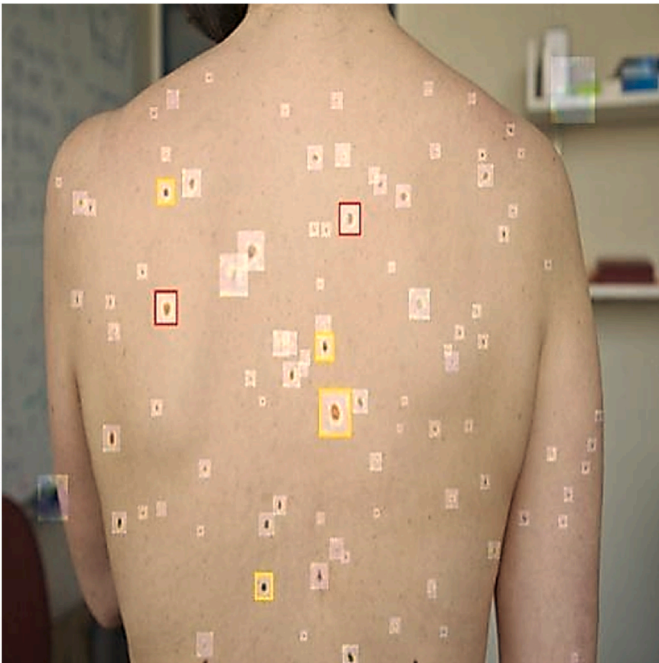


Fig. 17. Skin Cancer detecting app with the help of AI-assisted object detection models [159].

Most of the AI techniques and models, like the modern deep machine learning techniques that are trained to recognize and categorize skin cancer of the subjects or patients utilizing deep neural networks and computer algorithms-assist in diagnosing skin cancer have become a revolution.

There is more use of AI in the healthcare and medical sector globally, as shown in Fig. 18. The future of AI has more potential, and the prediction is within the next 30 years, they may even rapidly increase the number of preventions of death from any fatal disease.

The biggest challenge for AI in various medical services is not determining whether the innovation is strong enough to be helpful but ensuring its acceptance in clinical scheduling. The AI-based framework needs to be supported by controllers, coordinated with HER (Electronic Health Record) frameworks, normalized by enough people so that similar products function essentially in the same way, trained by doctors, and compensated by open or confidential payers to be widely used.

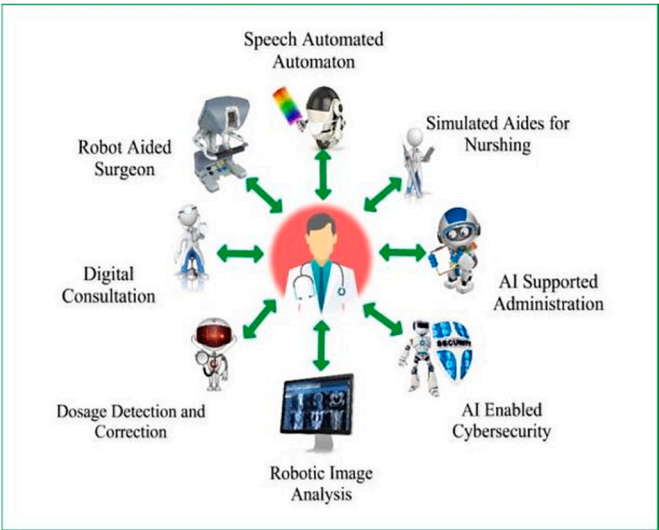


Fig. 18. Future of AI in the healthcare and medical sector development [162].

It should be purchased and put out long-term effort in the field. These problems will be resolved in the end, but innovation will take time and money to advance. And for the next five years, it is anticipated that there will be a limited application of artificial intelligence in clinical practice, followed by a wider one [139,278].

Some futuristic models, like Fig. 19, are already in development. The foundation is AI, which is likely to change the whole healthcare industry’s processing drastically. Seizure-detecting smart bracelets are another revolution in this sector. Waterproof wearable Epilept bracelets detect and track epileptic episodes using biometry and machine learning (Epilepsy Monitoring Unit). It is a smart wristband with Bluetooth functionality connected to a mobile application.

The integration of AI in healthcare revolutionizes diagnostic accuracy, treatment recommendations, and patient care. Through advancements like AI-assisted diagnostics and precision medicine, coupled with evolving technologies, the future promises transformative changes in healthcare delivery and patient.

4.5. AI in the autonomous systems

This might be the most intriguing and intimidating sector of AI. The market size of AI-driven autonomous systems was \$4.2 billion in 2022 [164]. 2023, it grew to \$4.8 billion [165], likely reaching \$59.54 billion by the end of 2030 [166]. Autonomous systems are widely used in transportation, with many consumer vehicles featuring assistive technologies but not fully autonomous. The Society of Automotive Engineers and the U.S. National Highway Traffic Safety Administration have defined a six-level classification system for vehicle automation, ranging from level zero (momentary driver assistance) to level five (fully automated with no human driver required). Presently, vehicles up to level

two are commercially available in the United States [167].

4.5.1. Worldwide use of AI in the autonomous sector

While self-driving cars often dominate headlines, AI’s application and implementation in autonomous systems, including vehicles, extends beyond them. Future advancements could see autonomous systems taking on tasks such as piloting aircraft, cargo delivery, and ambulance operations. Some researchers argue that AI could perform these tasks more efficiently and safely than humans. Moreover, advancements in autonomous technology are poised to unlock new opportunities in fields like medicine, scientific research, and robotic space exploration. The potential of AI in transportation extends far beyond self-driving cars, promising transformative impacts across various domains.

Autonomous vehicles have seen significant advancements, with technologies like long-range radar, cameras, and LIDAR playing pivotal roles in data collection. However, merely collecting data is insufficient without effective processing and deriving actionable insights.

Artificial intelligence (AI) is the brain behind autonomous vehicles, enabling them to interpret and act upon the collected data [31]. Some critical applications of AI in this domain include:

- **Automated Navigation:** AI algorithms can direct vehicles to gas or recharge stations when fuel levels are low, ensuring seamless operation without human intervention.
- **Dynamic Route Optimization:** By analyzing real-time traffic conditions and historical data, AI can adjust trip directions to identify the quickest route, enhancing efficiency and reducing travel time.
- **Enhanced Communication:** Incorporating speech recognition capabilities enables advanced communication between the vehicle and

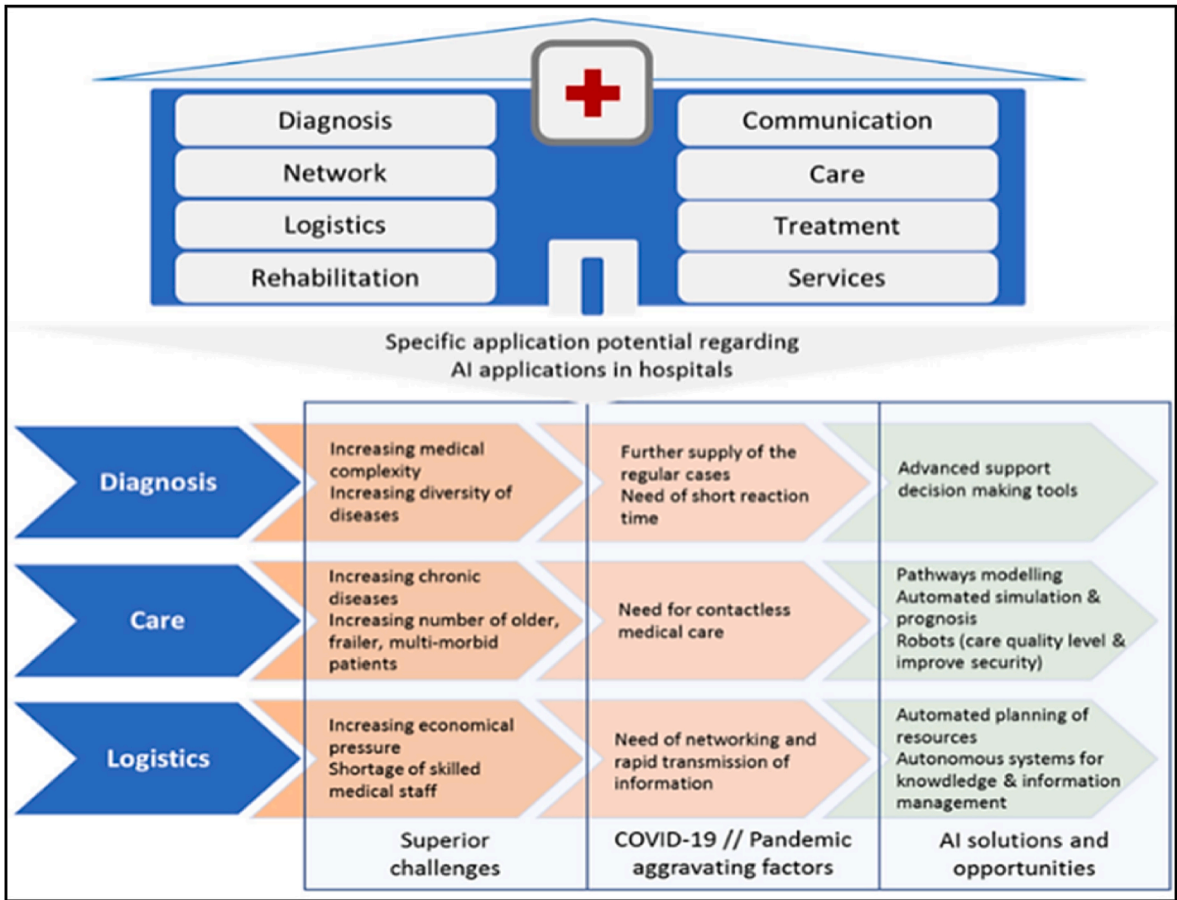


Fig. 19. AI assistance architecture for hospital administration-a futuristic plan [163].

its passengers, allowing for hands-free interaction and personalized experiences.

- **Natural Language Interfaces:** AI-driven natural language interfaces and virtual assistance technologies facilitate intuitive interactions between passengers and the vehicle's systems, enhancing user experience and accessibility.

4.5.2. Use of AI in forming different autonomous system based technologies

Table 6 highlights various AI-based autonomous systems spanning the transportation, military, and healthcare sectors. Examples include Waymo and Tesla Autopilot autonomous driving technologies, NVIDIA's Drive platform for vehicles, and military drones like Skydio X2D and General Atomics MQ-9 Reaper. Healthcare showcases predictive analytics (CLEW ICU EWS), clinical decision support (DeepMind Health Streams), and medical imaging analysis (Enlitic), all employing AI-driven autonomous systems.

Tesla's autopilot system utilizes a network of cars to gather visual data, as illustrated in Fig. 20, which is then fed into a training model for continuous learning. This data-driven approach enables real-time decision-making and ongoing improvements in autonomous driving capabilities.

Fig. 21 showcases SkyDio X2D drone, which was already mentioned in Table 6- one of the latest AI innovations and most adored drones among US soldiers. It offers intelligent decisions to avoid obstacles as integrated with the AI tech NVIDIA TX2.

Fig. 22 depicts the most advanced types of automated systems leveraging AI, including unmanned aerial vehicles, ground robots, and specialized systems for landmine detection, revolutionizing military operations. X-47B is an unmanned combat air vehicle developed by the United States Navy. It incorporates advanced AI for autonomous operations, including surveillance, reconnaissance, and strike missions. The TALON AI-based military robot is designed and manufactured by Foster-Miller. It is a versatile and rugged unmanned ground vehicle with artificial intelligence for various military applications. MQ-9 Reaper- A remotely piloted aircraft for long-endurance surveillance and strike missions. This Unmanned Aerial Vehicle (UAVs) utilizes advanced AI systems to enhance its capabilities in intelligence gathering and target engagement. Modern-day AI-augmented Landmine Detection System (ILDS) is developed to detect and neutralize landmines using artificial intelligence algorithms. The ILDS employs sophisticated sensors and AI technology to accurately locate and disarm landmines, ensuring safer environments in conflict zones. All of these were mentioned in Table 6, and this is their impact globally already.

Another ongoing AI integration in automotive systems is the M1 Abrams tanks' ATLAS AI targeting system. Fig. 23 showcases the US military and C5ISR engineers and scientists collaborating to test prototypes developed under the ATLAS program.

ATLAS utilizes advanced sensing technologies and machine learning algorithms [207] to expedite target acquisition and identification. Moreover, it recommends the type of weapons and ammunition to use. ATLAS enables crews to engage three times as many targets within the same timeframe. The system comprises optical sensors mounted on the tank or Infantry Fighting Vehicle (IFV) and internal PCs for image processing using AI. This necessitates additional cooling systems for the PCs, as depicted in one of the images.

Overall, AI plays a crucial role in enhancing the functionality and performance of autonomous systems, enabling them to navigate safely, efficiently, and intelligently in diverse environments. The AI integration is reshaping the whole sector and revolutionizing it.

4.6. AI in the telecommunications sector

Currently, the telecommunications sector (see Table 7) is a particular high-tech service sector. Tight business processes and IT application integration are the distinguishing characteristics of the telecoms sector. It has long been understood how important using IT is for boosting this

Table 6
Worldwide, the latest and most advanced AI-based tech products rapidly reshaping the autonomous systems sector.

SI No	Technology	Application/ Implementation	Company/ Organization	Source
1	Waymo	Autonomous driving technology: Waymo's self-driving technology uses AI algorithms and sensor fusion to enable vehicles to navigate roads, detect obstacles, and make real-time driving decisions.	Waymo (Alphabet Inc.)	[169]
2	Tesla Autopilot	Autonomous vehicle system: Tesla's Autopilot system combines AI-driven features such as adaptive cruise control, lane-keeping assistance, and self-parking capabilities to enable semi-autonomous driving.	Tesla, Inc.	[170,171]
3	NVIDIA Drive	Autonomous vehicle platform: NVIDIA's Drive platform offers AI-based software and hardware solutions for autonomous driving, including perception, mapping, planning, and control functionalities.	Various automotive manufacturers (e. g., Audi, Mercedes-Benz)	[172,173]
4	Argo AI	Autonomous vehicle technology: Argo AI develops AI-driven software and hardware systems for self-driving vehicles, focusing on perception, prediction, and decision-making capabilities to enable safe and efficient autonomous navigation.	Ford Motor Company, Volkswagen Group	[174]
5	Mobileye	Advanced driver-assistance systems (ADAS): Mobileye's ADAS technology integrates AI algorithms and vision-based sensors to provide lane-keeping assistance, forward collision warning, and pedestrian detection, enhancing vehicle safety and autonomy.	Various automotive manufacturers (e. g., BMW, Audi)	[175]
6	Skydio X2D	Autonomous drone technology: The Skydio X2D is a military-grade drone with AI-powered navigation and obstacle avoidance capabilities, enabling autonomous aerial surveillance and reconnaissance missions.	Skydio, Inc.	[176]
7	General Atomics MQ-9 Reaper	Unmanned combat aerial vehicle (UCAV): The MQ-9 Reaper, developed by General Atomics, is an AI-enabled UCAV used for	General Atomics Aeronautical Systems, Inc.	[177,178]

(continued on next page)

Table 6 (continued)

SI No	Technology	Application/ Implementation	Company/ Organization	Source
8	Taranis	reconnaissance, surveillance, and targeted strikes in military operations. Unmanned combat aerial vehicle (UCAV): Taranis, developed by BAE Systems, is an autonomous stealth drone equipped with advanced AI systems for conducting long-range strategic reconnaissance and strike missions.	BAE Systems	[179]
9	Titan Sentry	Autonomous ground vehicle: Titan Sentry is an AI-driven unmanned ground vehicle (UGV) designed for military applications such as perimeter security, reconnaissance, and convoy support, equipped with sensor arrays and autonomous navigation capabilities.	Milrem Robotics	[180,181]
10	Boeing Airpower Teaming System (ATS)	Autonomous aerial vehicle: The Boeing ATS, developed by Boeing Australia, is an AI-driven drone designed for intelligence gathering, surveillance, and reconnaissance missions. It is equipped with advanced sensors and autonomous capabilities, enhancing military situational awareness and operational effectiveness.	Boeing Australia (Boeing, 2022)	[182]
11	DARPA ALIAS	Autonomous helicopter technology: The DARPA Aircrew Labor In-Cockpit Automation System (ALIAS) program aims to develop AI-driven automation technologies to assist military helicopter pilots in various flight operations, including autonomous takeoff and landing, navigation, and emergency response.	Defense Advanced Research Projects Agency (DARPA)	[183,184]
12	DJI Phantom Series	Unmanned aerial vehicles (UAVs) for surveillance and reconnaissance: The DJI Phantom series offers AI-enhanced UAVs equipped with advanced imaging sensors and intelligent flight capabilities, enabling military forces to conduct high precision and efficiency in aerial	DJI	[185,186]

Table 6 (continued)

SI No	Technology	Application/ Implementation	Company/ Organization	Source
13	CLEW ICU Early Warning Score (EWS)	surveillance and reconnaissance missions. Predictive analytics for critical care: CLEW ICU EWS is an AI-powered early warning system deployed in intensive care units (ICUs) to predict patient deterioration and facilitate early intervention, improving patient outcomes and reducing mortality rates.	CLEW Medical	[187,188]
14	DeepMind Health Streams	Clinical decision support system: DeepMind Health Streams is an AI-powered mobile application designed to assist healthcare professionals in monitoring and managing patient data in real-time, enabling more informed clinical decisions and improving patient care.	DeepMind Technologies Ltd.	[189]
15	Northrop Grumman X-47B	Unmanned combat air vehicle (UCAV): The X-47B is an AI-controlled UCAV developed for autonomous takeoff, aerial refueling, and precision strike missions, demonstrating the potential for unmanned systems to operate alongside manned aircraft in combat environments.	Northrop Grumman Corporation	[190,191]
16	TALON Military Robot	Unmanned ground vehicle (UGV) for reconnaissance and bomb disposal: The TALON military robot is an AI-driven UGV designed for reconnaissance, surveillance, and bomb disposal tasks in hazardous environments, enhancing soldier safety and operational efficiency on the battlefield.	QinetiQ	[192,193]
17	Enlitic Medical Imaging	AI-based medical imaging analysis: Enlitic develops AI algorithms for medical imaging analysis, leveraging deep learning to assist radiologists in detecting and diagnosing various medical conditions from X-rays, MRIs, CT scans, and other imaging modalities.	Enlitic	[194]

(continued on next page)

Table 6 (continued)

SI No	Technology	Application/ Implementation	Company/ Organization	Source
18	Clinical Decision Support Systems (CDSS)	AI-driven decision support: CDSS integrates AI algorithms and medical knowledge databases to provide healthcare professionals with evidence-based recommendations for clinical diagnosis, treatment planning, and patient management, improving diagnostic accuracy and treatment outcomes.	Various healthcare technology providers	[195,196]
19	Lockheed Martin M1 Abrams	AI-enabled main battle tank: The M1 Abrams tank incorporates AI technology for advanced targeting, fire control, and battlefield awareness, enhancing its combat capabilities and survivability in modern warfare scenarios.	Lockheed Martin Corporation	[197,198]
20	Landmine Detection system	Landmine detection systems utilize various technologies such as ground-penetrating radar, electromagnetic induction, and chemical sensors to detect buried landmines and unexploded ordnance in conflict zones and post-war areas, improving civilian safety and facilitating landmine clearance efforts.	BAE Systems, Cobham PLC, and Schiebel Corporation.	[199–202]

industry sector's competitiveness. This is more of a reason why this sector has been implementing AI continuously for a long time.

The telecoms sector was the first to use artificial intelligence (AI) methods more than ten years ago. Macleish (1988) studied the connection between two fast-developing technologies, AI and telecommunications; simultaneously, the application of AI in telecoms started. At that time, expert systems created for diagnosing complex equipment offline were the primary usage of AI in telecommunications. In 1993, Muller emphasized the need for AI in telecommunications and offered a summary of AI techniques; Seshadri also included a brief discussion of AI in telecommunications. Since its early use, this sector has seen increased AI techniques [208].

These AI technologies are utilized by telecommunications companies worldwide to enhance customer service, automate network management, improve security, and enable innovative communication solutions.

4.6.1. Designing the telecom sector with AI

The telecoms sector is where most AI applications in network deployment are located. The design and topology are crucial for future FTTH (Fiber to the Home) point-to-point (P2P) communication network rollout. Some scientists suggest using AI to enhance this architecture. For instance, Martinelli, Andrioli, Castoldi, and Cerutti (2014) suggest a genetic algorithm-based AI technique to reduce the number of all-optical regenerators to reduce network costs in the telecommunications industry. Additionally, in the recent 21st century, several AI applications to support network operations have been identified in the telecoms sector [216].

For example, resource distribution in the backbone and network reconfiguration present a significant operational issue for telecom networks. AI supports the dynamic dimensioning of the backbone. The problem can be readily resolved by performing reconfigurations based on the traffic volume and direction forecasted by a neural network using ML. Numerous telecom companies already do that. Once more, AI applications in the telecom industry can help with IP network routing and connection establishment. While the sensors and transmission network must first be set up for smart electrical grids, telecom operators already have both in their current networks in many business scenarios. Therefore, it makes sense that some of the first applications of AI to have a market influence on network sectors do so in the telecommunications sector.

New AI-based technologies offered by telecom operators or their partners could search for patterns to detect, anticipate, and locate

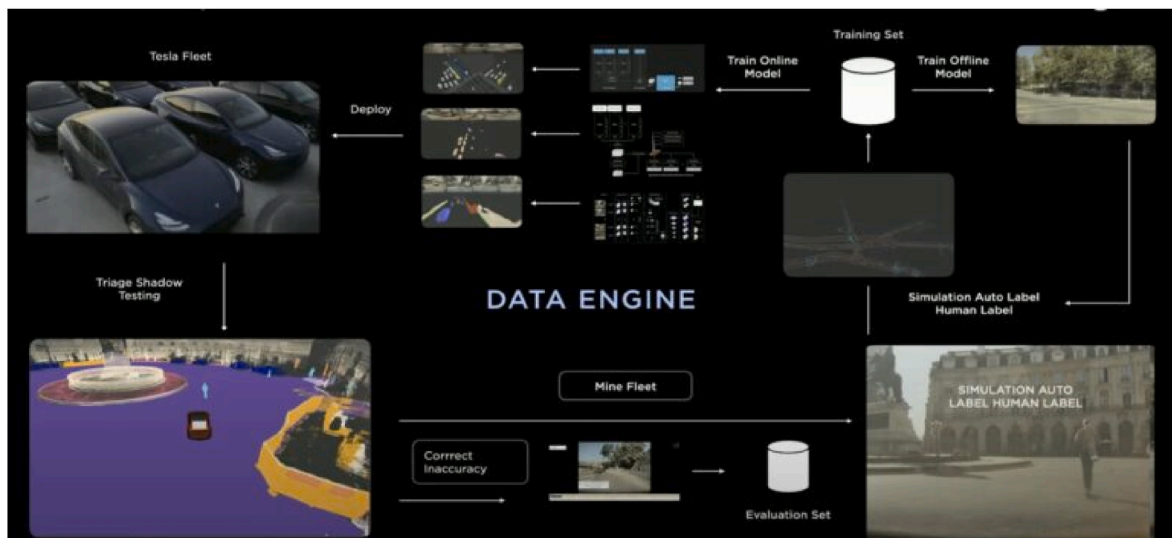


Fig. 20. Tesla's AI-integrated autopilot system (How it functions) [168].



Fig. 21. SkyDio X2D automated drone with AI-integration [203]. (a) SkyDio X2D drone 3D scanning. (b) AI-integrated 3D scan interface. (c) SkyDio X2D drone.

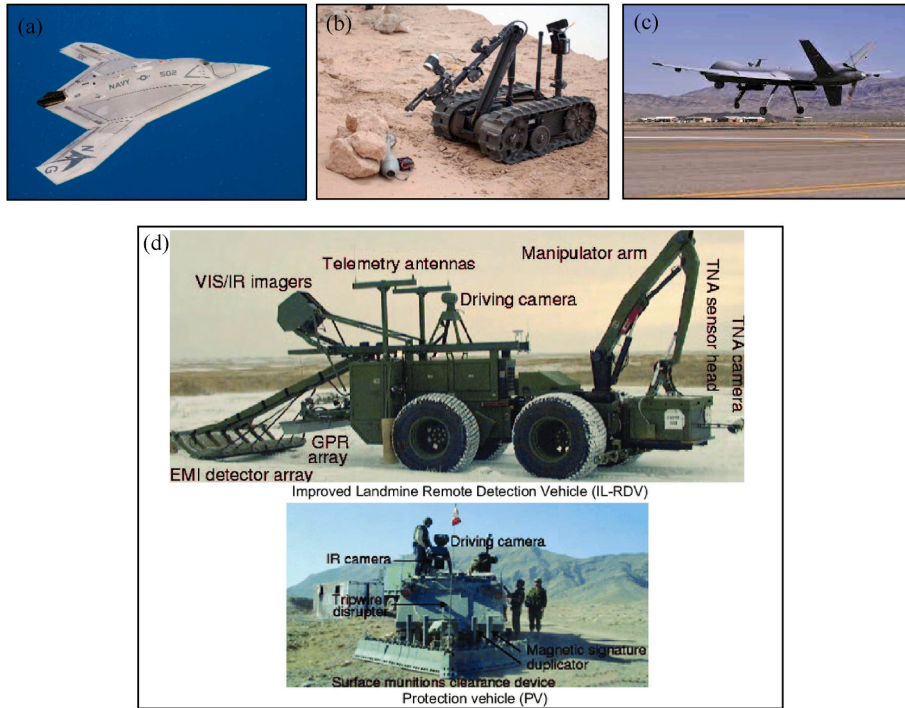


Fig. 22. Most advanced types of automated systems are based on AI and are helping advance the sector. (a) X-47B [204], (b) TALON ([205], (c) MQ-9 Reaper [206], (d) Modern-day AI-augmented Landmine Detection System (ILDS) [200].

network problems and immediately implement mitigation measures. To defend the network, they could, for instance, recognize a distributed denial of service (DDoS) attack, examine and identify pertinent package attributes, and discard all packages with those qualities [216].

4.6.2. AI-based drone technology for advanced communication

Avy Aera 3 VTOL Drone [217,218] is considered the best multi-modal surveillance drone, and Skeyetech by Azur drones [219] are the best 'No-training Drones' of this century. These AI-augmented drones are beneficial and efficient in maintaining surveillance in the telecommunication sector. Although the Avy Aera 3 VTOL Drone isn't intended to be a surveillance tool, situational awareness fits well into its mandate. The drone has been utilized for everything, from protecting wildlife to medical delivery. Winged drones are more energy-efficient and have a far more extended range than quadcopters. But compared to winged drones, quadcopters are significantly more agile. The best of both worlds is combined in Avy. Skeyetech by Azur Drones makes an alluring claim: It doesn't require security officers to have prior flying experience or pilot training. The drone can be deployed in under 30 seconds, and everything, including takeoff, landing, and course planning, is automated. Security teams use Azur's integrated video management system, which offers real-time HD footage to security Headquarters to request live operations or follow robotic missions.

The telecommunications sector has long been at the forefront of AI adoption, utilizing advanced technologies to enhance customer service,

network management, and security. From AI-powered virtual assistants to network automation, telecom companies leverage AI to stay competitive and innovative. Additionally, AI-augmented drones like Avy Aera 3 VTOL and Skeyetech by Azur Drones further contribute to surveillance and operational efficiency in the telecom industry. As AI continues to evolve, its integration into telecommunications promises even more significant advancements in network optimization, problem detection, and mitigation measures.

4.7. AI in the fashion and apparel sector

The fashion and apparel industry has seen significant change over the past few decades, much like other businesses, industries, and various other sectors across the globe. The fashion and apparel (F&A) industry, which alone contributes 38% to the whole economies of Asia-Pacific, 26% of Europe, and 22% of North America [220], reflects very well that, without any doubt, it is one of the major economic sectors in the whole world right now.

In Europe and the Asia Pacific, the predicted growth rates for the apparel market are 7.5% and 5.5%, respectively. Overproduction and product returns make this industry a major global waste generator, driven by consumer dissatisfaction with size, color, and style. To regulate eco-friendly production, the industry must become more customer-centric. AI is already helping make the sector more efficient and sustainable [221].



Fig. 23. The testing of the latest M1 Abrams with the AI integration (The ATLAS AI targeting system) [207].

Table 7
Mentionable AI-augmented Telecommunications tech and the companies using them.

SI No.	Technology	Application/ Implementation	Company/ Organization	Source
1	Conversica	AI-powered virtual assistants for customer engagement and sales	Vodafone, Orange, T-Mobile	[209]
2	OpenAI	Natural language processing, chatbots, language translation	Telefonica, AT&T, Deutsche Telekom	[210]
3	Pindrop	Voice authentication, fraud detection	Verizon, BT, Rogers	[211]
4	Ciena Blue Planet	AI-driven network management and automation	BT, CenturyLink, Telstra	[212,213]
5	Nuance Communications	Conversational AI, speech recognition, voice biometrics	T-Mobile, AT&T, Verizon	[213–215]

4.7.1. AI as a problem solver in apparel sector

Due to its ability to solve issues using traditional mathematical models, AI has drastically changed several industries over the past few decades, including health, telecommunications, transportation, and manufacturing, as we have already covered in-depth discussion and evaluation above of the application of AI in those sectors. These issues are dealt with using a variety of AI techniques, including neural networks (NN), Generalized Artificial linear models (GA), fuzzy logic (FL), evolution strategies, artificial immune systems (AIS), expert systems (ES), and multi-agent systems (MAS).

Numerous research outputs and revolutions demonstrate the utter superiority of AI triumph over conventional ways in light of various decision-making challenges [222]. Different governmental and privatized organizations are doing more research to increase the scope and implications of AI in this industry. Some of the research segments for AI

applications and implementation in the fashion and apparel industry are:

AI-augmented Apparel CAD systems to improve productivity and quality.

Apparel designs with the help of AI for the determination of structure, patterns, and design elements.

- Fabric-related issues solutions.
- Garment evaluation with AI assistance.
- Apparel manufacturing.
- Retailing option range.
- Apparel supply chain management.
- Marketing and content to reach out to the target audience.

Fig. 24, (a) Automated CAD systems: This image, courtesy of Optitex, depicts automated Computer-Aided Design (CAD) systems in apparel manufacturing [275]. CAD systems are software tools used in the fashion and textile industry to digitally create, modify, and visualize clothing designs. The image likely showcases the interface of CAD software or an example of a design being created or modified using such a system. (b) AI-augmented machinery in Apparel manufacturing: This image, courtesy of Apparel Resources, illustrates the integration of Artificial Intelligence (AI) with machinery in the apparel manufacturing industry. It likely showcases the use of AI-powered technologies, such as computer vision or machine learning algorithms, in automated processes related to garment production. The image may depict machinery or equipment equipped with AI capabilities, enhancing efficiency, accuracy, or productivity in the manufacturing process.

AI is already working globally in these segments of the apparel industry. However, little attention to these scopes would strengthen the apparel industry. Automating numerous AI instruments in material handling, spreading, cutting, and stitching can lower manufacturing costs and minimize errors [223].

Similar to how there are often opportunities for error in human work since it is challenging to consistently remember many such factors and apply the knowledge for precise property prediction. AI makes this possible since advances in processing and simulation have led to the definite and significant creation of various systems and techniques that can handle multiple variables. AI applications may now work with a wide variety of datasets during training to create a useful correlation between the variables and the properties of the products. As a result, during the past ten years, the usage of AI in the textile and apparel production industries has significantly increased for various applications [223,224].

4.8. AI in the global marketing

Artificial Intelligence (AI) revolutionizes marketing by enhancing customer experiences and streamlining processes. Companies can effectively promote their brands and products with remarkable precision through AI-driven algorithms and bots [273].

One notable application of AI in marketing is evident in online shopping experiences. Previously, searching for products without knowing their exact names could be frustrating. However, modern e-commerce platforms utilize AI-powered search engines that swiftly provide relevant results, almost as if they can anticipate consumer preferences. This level of personalization is exemplified by platforms like Netflix, which leverage AI to suggest movies and shows based on individual viewing history and preferences.

Netflix's predictive technology analyzes vast amounts of data to offer tailored recommendations, continually refining its suggestions as the dataset expands. This capability highlights the potential of AI to enhance user engagement and satisfaction. Looking ahead, AI's influence in marketing is poised to grow further. Innovations such as image recognition technology, exemplified by companies like CamFind, could enable consumers to purchase products simply by snapping photos. As

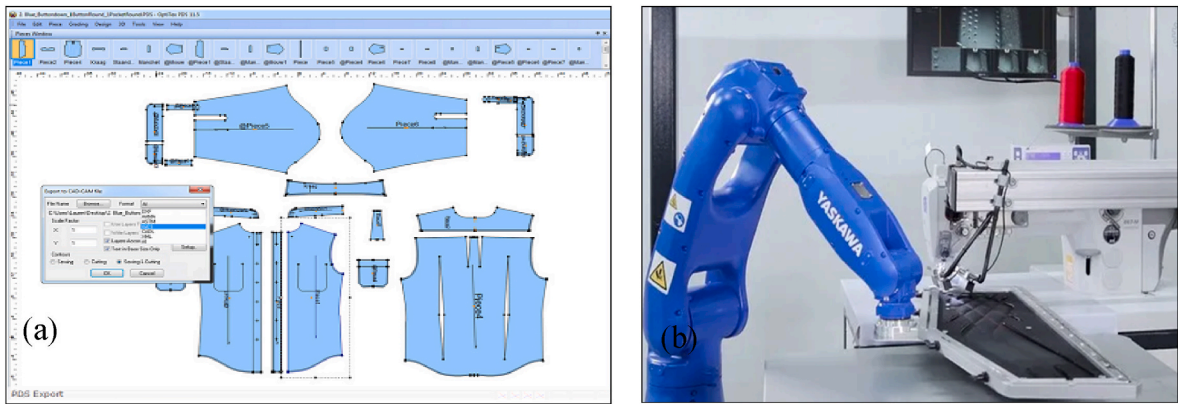


Fig. 24. (a) Automated CAD systems (Image courtesy: Optitex) (b) AI-augmented machinery in Apparel manufacturing (Image courtesy: Apparel Resources).

AI advances, it promises to revolutionize marketing strategies and redefine how companies interact with consumers online.

4.9. AI in the E-commerce sector

Amazon is the greatest example of the application of AI in the E-commerce sector. Amazon employs AI to provide services like voice shopping and product suggestions to users. Other applications of Amazon’s AI technologies include facial recognition, home price estimation, visual search, autonomous driving, and more. Amazon Echo and its built-in AI assistant, Alexa, allow customers to make voice-activated payments. The company’s checkout-free grocery store, Amazon Go, uses AI to identify selected items and monitor and extrapolate shopper behavior. Chicago, New York, San Francisco, and Seattle are active locales.

Although there is more demand, business AI integration is still tricky. According to a recent statistical poll, 85 % of AI programs eventually fall short of their goals [225]. Risk management is increasing due to the rise in e-commerce orders to prevent payment default [226]. At the same time, retail and e-commerce companies like Clarifai, eBay, Anaplan, IBM, Emotive.io, inVia Robotics, Alibaba, and many others are shining with the help of AI.

AI-based picture and video identification from Clarifai assists in classifying photographs and removing undesirable visual information. Human moderators can be added to the mix for an additional level of accuracy when analyzing other data that can go by AI’s radar. eBay uses AI to help customers, speed up shipping and delivery, lower prices, build trust between buyers and sellers, and do much more. Using real-time statistical data and accurate information that more precisely reflects a consumer’s dynamic purchase status- IBM’s Watson employs AI to assist retail organizations in creating more tailored shopping experiences. Emotive enables e-commerce businesses to communicate with customers through interactive conversational marketing, text messages, SMS payments, and customer support. Users can personalize their marketing messages and monitor relevant engagement data on the company’s Emotive Platform.

By the end of 2023, 73 % of eCommerce transactions are anticipated to be completed on mobile devices. Customers will, therefore, enjoy a more seamless mobile user experience, enabling them to buy just as simply on their phones as they can on their computers. Applications like Shopify, Square, WooCommerce, ShipStation, ThriveCart, eBay, Kajabi, and many others have impacted consumers. There are many more examples of the application of AI worldwide and success with their business strategy depending on AI. But, acknowledging their challenges and working to resolve those would strengthen the AI sector.

4.10. AI in the banking sector

AI is rapidly transforming the banking sector (see Table 8), with many institutions leveraging AI-based systems for customer support, anomaly detection, and fraud prevention. A prime example is HDFC Bank, which has introduced EVA (Electronic Virtual Assistant), an AI-powered chatbot developed by Senseforth AI Research.

Since its launch, EVA has efficiently handled over 3 million customer queries, engaged with more than half a million unique users, and conducted over a million conversations. With the ability to gather insights from numerous sources, EVA delivers prompt responses in less than 0.4 seconds, enhancing customer service.

Fraud prevention is another critical area where AI excels. AI algorithms effectively identify and prevent fraudulent activities by analyzing transactional behaviors and access patterns. Major players in the banking industry, such as MasterCard and RBS WorldPay, have long relied on AI and deep learning techniques to detect fraudulent transaction patterns, resulting in substantial cost savings and heightened security measures.

These AI technologies are crucial in modern banking operations by enabling personalized customer interactions, improving risk management, enhancing cybersecurity defenses, and combating financial crimes.

Table 8
Application of AI in the field of banking worldwide.

SI No.	Technology	Application/ Implementation	Company/ Organization	Sources
1	Kasisto	Virtual banking assistants	JPMorgan Chase, DBS Bank	[227]
2	Quantexa	Financial crime detection and fraud analytics	HSBC, Standard Chartered	[228]
3	Personetics	Personalized customer engagement	Citibank, RBC, Santander	[229]
4	Darktrace	Cybersecurity for banking networks	Barclays, Lloyds Banking Group	[230]
5	Feedzai	Fraud detection and prevention	Capital One, Barclays	[231]
6	Kofax	Document automation and processing	Bank of America, HSBC	[232]
7	ZestFinance	Credit underwriting and risk assessment	Goldman Sachs, Discover Financial Services	[233]
8	Mambu	Cloud banking platform	Banco del Estado de Chile	[234]
9	Temenos	Core banking software	Banco de Crédito del Perú	[235]
10	Tinkoff Bank	Digital banking services	Tinkoff Bank (Russia)	[236]

4.11. AI in the transportation sector

The transportation sector is skyrocketing now, a credit to AI technologies [237–239]. The transportation sector is one of them. The ANN (Artificial Neural Network), GA (Genetic Algorithm), SA (Simulated Annealing), AIS (Artificial Immune System), ACO (Ant Colony Optimizer), BCO (Bee Colony Optimizer), and fuzzy logic models are examples of artificial intelligence technology discovery [240].

Road traffic management uses artificial intelligence (AI) tools to analyze the volume, pattern, and other aspects of traffic. To help with any traffic congestion that may have developed, these models can then give drivers information on the quickest path. Artificial neural networks (ANN), depicted in Fig. 25 (a), are being utilized to determine the position of vehicles on the road and to estimate their speeds. Also, other AI models like GA (a), SA (b), and FLM (c) architecture are illustrated in Fig. 26, which are continuously in use for the making of Intelligent Transport Systems worldwide. In most conventional Intelligent Transport Systems, route forecasting and destination planning based on prior travels (ITS) are missing components.

A transportation problem can be optimized using a genetic algorithm. However, the method takes significant processing resources compared to the paper-and-pen methodology. When the search space is ample and the population contains a variety of fitness factors, this strategy can be advantageous. A transportation problem can be optimized using a genetic algorithm. However, the method takes significant processing resources compared to the paper-and-pen methodology.

When the search space is ample and the population contains a variety of fitness factors, this strategy can be advantageous. Using the model, for instance, to produce a daily truck timetable for delivering a list of requests in the most optimized and efficient ways. Several fields, including transportation engineering, have used fuzzy logic. It has been used in transportation engineering to simulate traffic flow, transportation planning, and traffic control, among other things like planning for transportation in general, traffic management, etc.

The smart city is the best example of an application of AI that indulges the idea of a city with the expansion of digital and cloud-based technology, undefined innovative strategies, and strategic decision-making that is sustainable, integrative, knowledge-generating, and people-focused [243]. With AI, an intelligent city integrates information and communication technologies (ICT) to cut expenses, optimize resource usage, boost interaction, and improve residents' quality of life.

4.12. AI in the entertainment and gaming sector

AI was initiated in this sector, as discussed in 'the history of AI'-Games. This sector alone is the biggest commercial side of AI. Artificial intelligence has a long history of using game applications in the entertainment zone. Games are processed in a constrained domain, offer a wide variety and scalability for issue formulations, and the outcomes are typically simple to analyze. However, there is also a lot of business interest, as the "AI inside" aspect is a top priority in the rapidly expanding, multi-billion-dollar electronic gaming market. The revenue from PC

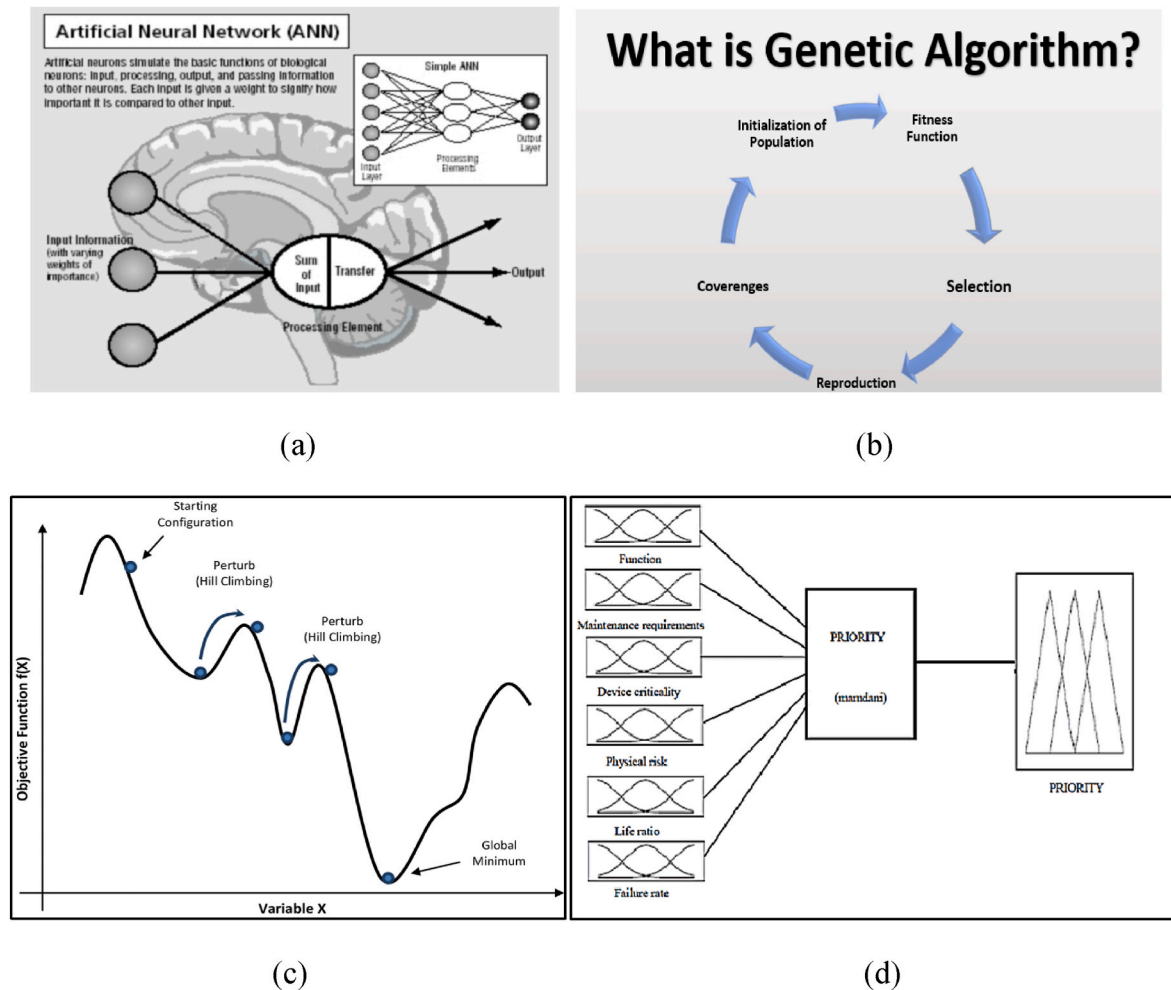


Fig. 25. Different prominent ML algorithms which are a component of AI that are being used in the Transportation sectors - (a) Artificial Neural Networks ANN (Image courtesy: Civildaily) (b) Genetic Algorithms (GA) (Image courtesy: Educba) (c) Simulated Annealing (SA) [241] (d) Fuzzy Logic Model [242].

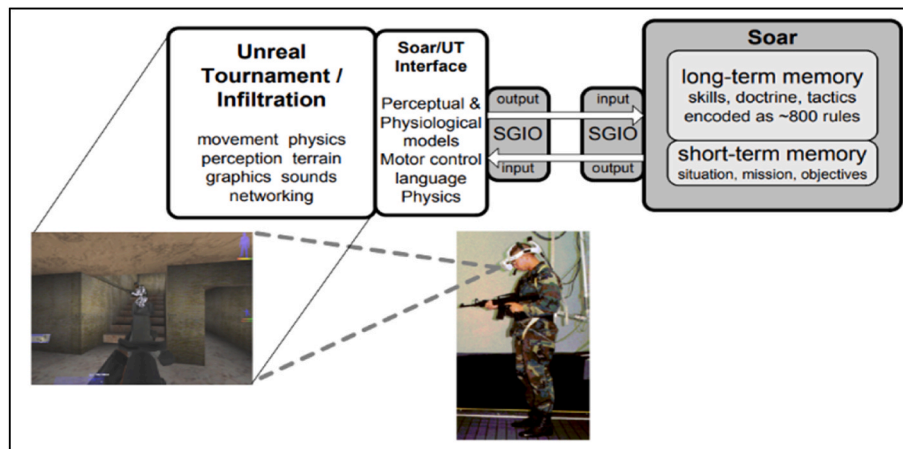


Fig. 26. The MOUTbot system architecture (used to enhance military proficiency).

game software created by AI and ML is already greater than that of box office hits [244].

AI approaches have lately been used to solve many "traditional" games, including Go-Moku and the Nine Men's Morris, as well as card, board, and puzzle games. Another critical moment for AI was when Deep Blue defeated Kasparov. Computer games have been dubbed the "Killer Application for Human-level AI." Even though most games are more complex now with modern dynamics, Real-time action graphics, Incomplete knowledge fillers, Resources, and animation, most games carry the primary type of AI techniques.

The desire to use AI methods, particularly CBR approaches, in computer games has significantly increased in recent years. At ICCBR 2005 and IJCAI 2005, seminars and workshops were explicitly devoted to game AI [245].

But there are still some challenges in this sector regarding the implementation of AI, like:

- Complex in-game design space.
- Knowledge engineering on authoring the behavioral sets.
- Authoring support in the game.
- User-specific adaptation.
- Unanticipated situations and encounters of various player strategies.
- Rhetorical objectives.

Overcoming these would pave the way for smoother AI and help organizations make more financial generation.

These were some of the biggest sectors in which AI dominates and influences the most. Aside from these, there are also other perspectives of AI where it has a major impact. One is image processing with object detection and recognition techniques with AI augmentation. This is also under the automation sector. A computer vision approach called object detection and recognition is used to locate items in pictures and videos. It is a crucial result of machine learning and deep learning algorithms. Automated object detection's core aim, objective, and major goal are to support machine learning and teach what comes naturally to humans.

The pinnacle of today's modern culture is the capacity to quickly identify objects in any video or image, such as animals, insects, humans, cars, animals, things, or substances, and more. After AI has acquired a level of comprehension of what a picture consists of because of receiving these teachings, it will be able to analyze and highlight specific aspects of an image. The following chapter will discuss More details on this subject matter.

With 188.2 million digital gamers in 2019 and 201.9 million gamers in 2020, in 2023, there will be over 3 billion gamers. The numbers reflect how fast this industry has grown and is likely to increase. Due to its extreme popularity, AI-augmented gaming app testing has gained

more significance in guaranteeing perfect experiences for players.

NPCs (Non-Players Characters) have been one of the greatest examples in the gaming industry for a long time. Any significant automated issue or strategy we encounter in the design of video games is called artificial intelligence, encompassing all incarnations of adaptability and responsiveness in games. Everything in the game is an AI setup, from conception to setting, launch, and play. Even your characters are built upon the function of AI, though their response is in the player's hands of choice. Most of these AI-powered interactive experiences are typically made by NPCs, or non-player characters, who act intelligently or imaginatively as though a human gamer was controlling them. AI controls an NPC's behavior in the game's universe. The Merchant of Resident Evil 4 game, Cortana of HALO, Errors of Zelda II, etc., are the most outstanding examples of NPCs in gaming history.

Copyright infringement has increased with the growth of social media, which is a major issue for the media and the music business. Currently, AI aids in the detection and prevention of copyright infringement just like SAFE by CodeMatch [246]. YouTube uses Google's machine learning technology, the biggest and most popular video streaming network on the planet, to enforce AI-enhanced copyright. But there are some far better options for the detection of copyright infringement, like SAFE's (Software Analysis and Forensic Engineering) CodeMatch tool (Fig. 27), which has the working principle of completing the assignment by comparing the source code of each file in the first set with every file of code in the second set. The relative quantity of matching code in the files is used to sort the results table that displays the results.

In conclusion, the entertainment and gaming sector is a major commercial domain for AI, significantly contributing to revenue and technological advancements. Challenges persist, including complex game design and user-specific adaptation, but AI continues to shape gaming experiences and address copyright infringement issues with tools like SAFE's CodeMatch. AI's role in gaming and entertainment will expand as the industry evolves, ensuring immersive experiences and protecting intellectual property rights.

4.13. AI in the military and Homeland security

AI has important uses in many fields, and the military is no exception. Incorporating AI technology into military operations has greatly improved efficiency, decision-making, and overall strategic advantage. This section discusses AI's potential benefits and applications in the military [277].

Artificial intelligence (AI) systems and technology are crucial to enhance military proficiency in various domains. The military's primary application of AI is in autonomous weapon systems. These technologies

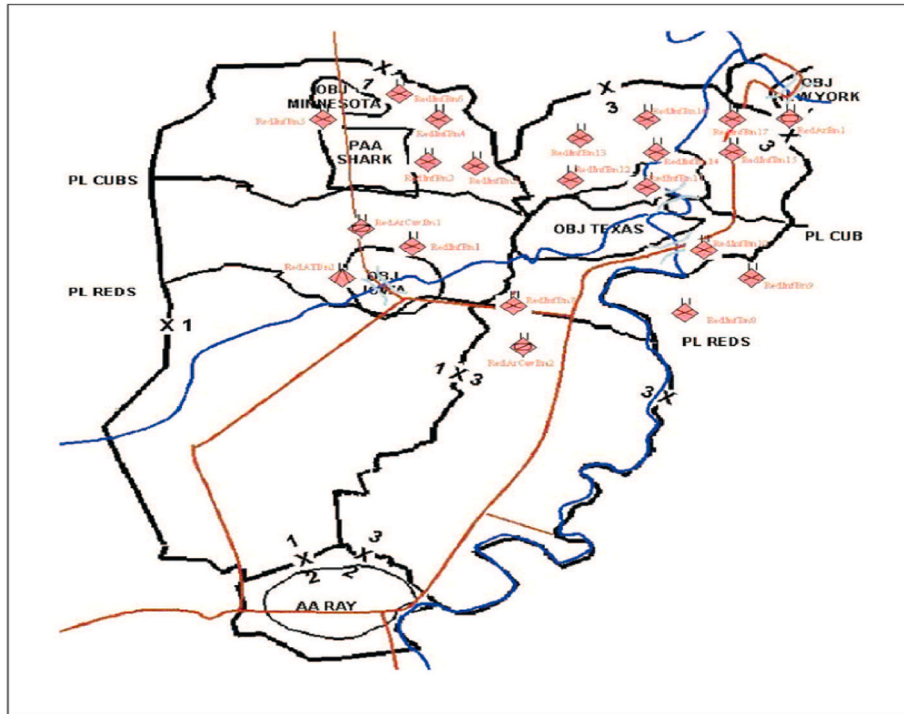


Fig. 27. A course-of-action sketch used in one of the US military experiments based on decision-making AI analytics. Note the sheer number of complex relationships between the significant features such as friendly and enemy units, roads, assembly areas, objectives, and axes of attacks [248].

eliminate the need for human involvement in warfare by automatically locating, tracking, and attacking their targets using sophisticated AI algorithms. For instance, autonomous drones can be used for surveillance, reconnaissance, and even precise target killing.

Artificial intelligence (AI) is also useful in simplifying combat assessments and choices. Machine learning algorithms can sift through real-time data gathered from sensors, satellites, and intelligence databases. This skill enables military leaders to make quick decisions based on accurate data, adjust to new circumstances, and anticipate potential dangers [247].

Fig. 26 depicts the system architecture of the MOUTbot (Military Operations in Urban Terrain robot), a system designed to enhance military proficiency in urban environments. The image likely showcases the various components, modules, or subsystems of the MOUTbot system, highlighting its overall structure and functionality in military proficiency.

For a better understanding of decision-making AI, the below Fig. 27 can help understand pretty clearly how the US military uses AI to find a better routing for their vehicles and maintain a standard strategy for warfare pathway finding:

Advanced missile defense [249] and tracking systems would not be viable without AI as their decision-making is enhanced with AI's help, as

illustrated in Fig. 27. These systems use artificial intelligence algorithms to detect and neutralize potential dangers. As a result, defenses will be stronger overall. Artificial intelligence (AI)-driven predictive analytics models can also aid in risk assessment and mitigation, task success, and casualty avoidance.

The military also makes effective use of AI for cyber defense. Artificial intelligence systems may examine network traffic, detect anomalies, and identify potential cyber threats as the military increasingly relies on interconnected networks and digital infrastructure. Artificial intelligence (AI)-based cybersecurity systems can swiftly respond to threats, safeguard sensitive data, and defend vital military assets [250].

While AI holds great promise for military applications (Fig. 28), ethical considerations must be given due diligence. Responsible implementation and adherence to international laws and standards are essential for keeping AI-enabled systems accountable and minimizing the risk of unexpected outcomes.

AI is even aiding the military health system. One of the significant uses of AI-augmented robotics is extracting injured individuals from the battlefield or other areas that humans can't easily access.

AI has altered military strategy and the construction of defenses. AI technologies fundamentally reshape military operations from autonomous weapon systems and improved decision-making to cyberdefense

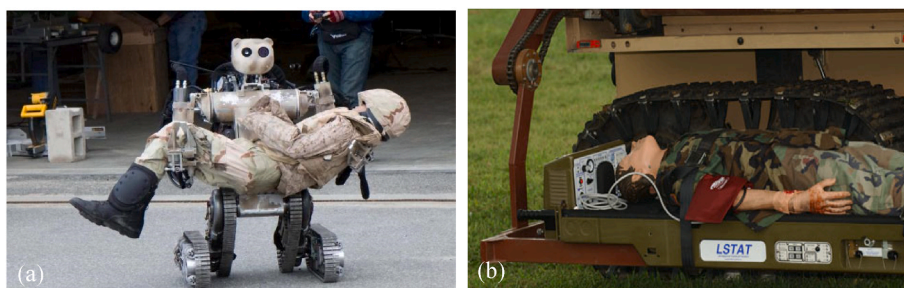


Fig. 28. (a) Battlefield extraction assist robot (BEAR) (b) Life support for trauma and transport (L-STAT) integrated with REV [251].

measures. It is crucial to balance technical advancement and ethical issues as AI improves. This will guarantee that the military’s access to AI’s benefits is used ethically and for the common good.

5. The future of AI, challenges, and threats

Recent news stories of the engineering world is now revolving around AI and its possible benefits, challenges, risks, threats, and results (both wins and failures). What does its near-term prospects look like? Having the benefit of reflection makes it clear that the best predictions can be made after the fact. If all kinds of lying are wrong, intelligent players will bet on things that can’t be proven or disproven. On the side of things that are not controversial, you could say that lawmakers are putting more pressure on AI developers to ensure their apps and inventions follow socially acceptable rules. For example, everyone expects the EU to take legal action at some point [252]. On the other hand, some people will keep spreading catastrophic predictions about dystopian futures that will happen so far in the future that the Jeremiahs won’t be around to see how wrong they are. Movies about vampires and the dead do well at the box office, so expect more of them. "Look into the seeds of time and say which grain will grow and which will not" (Macbeth, Act I, Scene III) is hard to do, and it could be embarrassing in the future if anyone tries to do it. This means figuring out where AI is more likely to go and, by extension, where it might not go.

5.1. The future of AI

The future of AI is undoubtedly bright, with endless possibilities for innovation and progress. The exponential development of AI technology over the past few years has been nothing short of astonishing, and the prospects for its future development are even more intriguing. Artificial intelligence is anticipated to revolutionize various fields soon and substantially affect our day-to-day lives [253].

AI holds immense potential to revolutionize various sectors in the near future than it already has including healthcare, finance, transportation, and other industries. In healthcare, AI promises early disease diagnosis, personalized patient care, and support for medical professionals in diagnosis and treatment planning. Similarly, AI can assist in fraud detection, stock market predictions, operational automation in finance, and many more.

Moreover, AI-driven self-driving cars aim to reduce accidents, traffic congestion, and transportation costs, promising enhanced efficiency. AI is poised to significantly impact healthcare, offering tailored treatment regimens based on genetic data and medical history, leading to more effective interventions and improved patient outcomes. In an overall expansion, we can see a future of AI in almost every industrial aspects.

5.2. Challenges of AI

However, the future of AI that we are discussing in this paper, after thorough research, is not guaranteed by any means; it requires scholars, legislators, and business executives to devote a tremendous amount of time and resources to this endeavor. As artificial intelligence (AI) continues to advance and become more ingrained in our daily lives, we must confront the problems and potential dangers it poses. We can only ensure that artificial intelligence is developed and deployed in a manner that is responsible, ethical, and beneficial to all of humanity if we work together [25]. To ensure that AI will continue to transform industries and improve our lives in the future, we must analyze and address the problems enumerated in Table 2 through a collaborative effort involving the government, academia, and the private sector.

Table 10 outlines the issues currently confronting AI, as well as some of the potential solutions (see Table 9). These issues must be resolved for AI to have a promising future. These obstacles include concerns about bias and ethics and the need for ongoing investment in developing artificial intelligence systems. In general, the emergence of AI is

Table 9
Potential Future Applications of AI in different Industries.

Industry	Potential Applications of AI	Impact
Healthcare	Personalized medicine, disease diagnosis, drug development	Improved patient outcomes, more effective treatments
Automotive	Self-driving cars, traffic optimization	Safer roads, reduced traffic congestion, lower emissions
Retail	Personalized shopping experiences, inventory management	Increased sales, more efficient operations
Finance	Fraud detection, risk management	Improved security, more efficient operations
Education	Personalized learning, automated grading	Improved educational outcomes, more efficient operations
Manufacturing	Predictive maintenance, quality control	Reduced downtime, improved product quality, lower costs
Agriculture	Precision farming, crop monitoring	Increased yields, reduced waste, more efficient operations
Energy and Utilities	Predictive maintenance, grid optimization	More reliable and efficient energy production and distribution

Table 10
Potential challenges and solutions for the future of AI [254].

Challenges	Potential Solutions
Bias in AI models	Improved data collection and algorithm design
Lack of transparency	Increased transparency and explainability in AI
Job displacement	Investment in education and job training programs
Ethical concerns	Development of ethical frameworks and regulations
Security concerns	Development of more secure AI systems
Data privacy	Increased data privacy regulations and standards
Regulation and control	Development of regulatory frameworks for AI

associated with several obstacles and potential hazards; however, there are many potential benefits. If we confront these issues head-on, we can help ensure that artificial intelligence continues to drive development and innovation across various industries and improve people’s lives everywhere.

Aside from these, some other common challenges of AI and its future could be:

- **Data Privacy and Security:** One of AI’s significant challenges is ensuring data privacy and security. As AI systems require large amounts of data to operate, there is a risk that this data could be misused or stolen, leading to a breach of privacy or security [255].
- **Bias, Fairness, and Ethics:** Biasness, fairness, and ethics of AI technologies and AI systems are not clarified enough, or the ways they work are not adequately understood. The technology or the system may function well, rightfully and ethically; only the data and the information given to the AI to train with are as unbiased as they should be for transparent decision-making. If the data is biased, the AI system will also be limited. This can lead to unfair outcomes, such as discrimination in hiring or lending [256].
- **Lack of Transparency:** AI systems are often considered black boxes [257]. The idea or logic of how they arrive at their decision or come to solve a problem is not clear yet to the researchers. This challenge is defined as a lack of transparency, which can easily correct errors or biases in the system. For a better illustration of this challenge, a retail store located in Detroit had Shinola watches stolen, and the local police were attempting to determine who was responsible. According to the authorities, the thief stole approximately \$3800 worth of property before escaping the scene. Investigators retrieved the video surveillance system that had captured the incident. After zooming in on the unsteady footage, the detectives utilized face recognition software to identify the suspect. The investigators retrieved the video

surveillance system that had captured the incident. After zooming in on the shaky footage, the detectives used face recognition software on the person who appeared to be the culprit. Unfortunately, the hit that came through the AI system later found that person innocent. So that's how a well-trained AI leads to a catastrophe and does not show any cause or characteristics on why it happened or what has gone wrong [258].

- **Lack of Skilled Professionals:** There is a shortage of skilled professionals who can develop and implement AI systems in industrial setup. This shortage could limit the development of AI and slow its adoption [254].
- **Existential Risk:** Some experts have raised concerns about the potential existential risks of advanced AI, such as an AI system becoming too powerful or uncontrollable.

To overcome these challenges, the world needs to focus on developing solutions that address these issues. This could involve increasing investment in research and development, improving data privacy and security laws, and promoting transparency in AI systems.

5.3. Threats of AI

It is vital to consider both the possible benefits and the potential threats as the field of artificial intelligence, AI, continues to develop and evolve. Nobody knows if artificial intelligence will help us get more intelligent, as Google's Raymond Kurzweil thinks it will, or if it will lead to World War III, as Elon Musk fears. On the other hand, most people agree that this will bring up new social, legal, and philosophical questions that need to be answered. The "Trolley Problem" has been a source of debate in the field of ethics for a long time.

In this imaginary situation, a person must choose between doing nothing and causing the deaths of many people or doing something and causing the deaths of only a few [259]. In a future where cars drive themselves, these choices will have to be made by robots and, by extension, by the people who program those machines. As a clear response, many people, including well-known people like Mark Zuckerberg, have asked for rules to be implemented. But it would be hard to deal with these risks and problems if they were caused by a technology that is constantly changing on its own and that only a few experts, let alone politicians, fully understand. Politicians are the ones who make decisions about the growth of AI [25].

It is impossible to overlook AI's impacts on society, industry, and the environment. This chapter will address the numerous hazards, impacts, and threats that artificial intelligence poses. We need to deal with these dangers head-on and develop solutions if we want artificial intelligence to have a bright and long-lasting future.

Table 11 lists the most common and recent AI threats and mitigation strategies. These threats include job displacement, security and privacy risks, autonomous weaponry, malicious use, environmental effects, and superintelligence. Stakeholders must act to prevent these hazards. This may include investing in education and training programs to help

workers adapt to new jobs, strengthening cybersecurity to protect sensitive data, and banning autonomous weaponry [261]. Responsible AI development and deployment should include ethical frameworks and interdisciplinary teams to examine AI's possible social impact. Let's dive in more deeply to explore these threats of AI:

Job Displacement and the resurgence of unemployment globally: One of the biggest threats of AI is job displacement. As AI systems become more advanced, they have the potential to automate many jobs, leading to widespread unemployment. As AI technologies evolve, employment concerns grow. AI can automate many human tasks. AI could displace numerous jobs, causing widespread unemployment (In the current worldwide economic market climate, it already has). Especially in automating repetitious work, low-skilled professions are most likely. For instance, faster, more efficient robots may replace manufacturing assembly line personnel. AI can automate data entry and record keeping, which may eliminate office jobs. AI will displace many jobs. McKinsey Global Institute estimates that 800 million jobs could be automated by 2030 [262]. This might upset economies and societies, especially in low-skilled labour countries.

Security and privacy risks: AI models need data to train and improve. Personal, financial, and commercial data may be in this data. Hacking or stealing this data could pose major security and privacy hazards for individuals and organizations. Hackers and cybercriminals could exploit this data for targeted assaults like phishing or identity theft. They could use the info to hack systems and networks and wreak severe damage. AI systems may also be hacked. If this happens, the system may make wrong or hazardous conclusions. For example, an autonomous vehicle could be hacked, causing it to crash, or a medical AI system could be hacked, leading to incorrect diagnoses or treatment recommendations that will be hazardous beyond our imagination [263].

Malicious Use: hackers or other bad actors could use AI systems maliciously to launch cyber-attacks, spread disinformation, or engage in other nefarious activities [264]. One of the biggest worries about AI is that hackers and other bad people might use it in bad ways. As AI systems get more innovative and more advanced, they might be able to start cyber-attacks, change data, and cause much trouble. Deepfake technology, which uses AI to make fake videos and audio that sound and look real, could be used to spread fake news and propaganda. In the same way, bots driven by AI can be used to make fake social media accounts, change the way people talk about politics, and spread fake news. This seriously threatens the security of democracy and the country [265].

Autonomous Weapons System (AWS) threat: AI-powered autonomous weapons could be developed and used to engage in warfare without human intervention, leading to significant ethical and moral concerns. This is one of the major threats to AI that the world is already concerned about more than anything. It alone could lead to global destruction and instigate global instability or chaos. AI-powered autonomous weapons use data to make judgments without human input. This technology could evolve into fatal weapons employed without human supervision, raising ethical and moral considerations.

The United Nations Conference on Disarmament talked about automated weapons in 2020 and called for a ban on them before they are made. This was done because people worried that countries could make weapons that could fight wars without human help. In 2018, a Turkish company, STM, made an autonomous drone that killed a person during a test run. Even though the drones were controlled from a distance, they could find and attack the target independently. This event raised concerns about how autonomous weapons could be misused [266]. In 2017, the Convention on Certain Conventional Weapons (CCW) of the United Nations held a meeting about weapons that can act independently [267]. During the meeting, many experts and groups discussed how worried they were about developing these weapons and the risks they could cause. In 2015, AI experts and researchers wrote an open letter asking for a ban on self-driving guns. The letter talked about how dangerous these weapons could be and asked that international rules be made to stop them from being made or used. These are just some recent

Table 11

Threats of AI and potential solutions [260].

Threat	Potential Solutions
Job Displacement	Education and training for new job skills
Malicious Use	Robust cybersecurity measures and ethical guidelines
Autonomous Weapons	International agreements and regulations banning their use
Superintelligence	Responsible development and regulation of AI technologies
Inequality and Bias	Diverse and representative data and ethical guidelines
Surveillance and Control	Strong privacy regulations and oversight mechanisms
Lack of Accountability	Transparent decision-making processes and human oversight

examples that show how dangerous autonomous weapons can be. If history is dug deeper, there could be some more rough real-life incidents that people might not like or expect. The level of threat these AI-augmented AWS could generate and the importance of international rules to ensure they are made and used responsibly and ethically. If not, there will be consequences that mankind doesn't want to face [268].

Superintelligence: If AI systems become superintelligent, they could potentially become a threat to humanity; this threat has been long predicted from the beginning era of AI. Some consider this threat superficial or fictional [269], while most expect this every time a new AI system develops and emerges on the face of the earth. This is because they could develop goals and values that conflict with human values, leading to unpredictable and potentially dangerous outcomes [270].

Environmental threats and impacts: The creation and use of AI technologies could affect a world that was not planned. For example, the massive amount of data AI needs to process takes a lot of energy and computing power, leading to more carbon emissions and energy use. Researchers at the University of Massachusetts Amherst did a study and found that teaching a single AI model can release as much carbon as five cars over their entire lives (Per the information of www.technologyreview.com). As AI systems get more complicated and used by more people, their environmental effects will likely worsen. Developing sustainable AI solutions and strategies to reduce energy use and carbon pollution is essential.

Some scholars might say these threats are inevitable-if AI grows further, humanity will likely face these threats and impacts. So, it is best to be concerned with these statements of threats and all other possible threats of AI. Before initiating the development of an AI system or technology, let's evaluate the consequences first. To overcome these threats, it is essential to develop policies and regulations that govern the development and use of AI. These policies and regulations should ensure that AI is developed and used to benefit humanity rather than detrimentally.

Investing in research exploring AI's ethical and social implications and promoting public awareness and education is essential. In general, the advancement and implementation of AI technology present a mixed bag of opportunities and difficulties. We will be able to reap the benefits of AI while minimizing the possible threats it poses if we take a proactive approach and collaborate.

6. Conclusion

As we approach the conclusion of this review paper, it has become abundantly evident that artificial intelligence has emerged as a disruptive technology that has far-reaching ramifications for various business sectors. Throughout this paper, we have investigated the inner workings of artificial intelligence (AI) and its myriad applications, including anything from speech recognition to image processing, natural language processing to robotics, and much more.

This review paper presents the findings of an in-depth investigation conducted on the current level of AI technology. The purpose of this study was to highlight how AI has the potential to alter organizations all over the world.

The research's contributions can be summarized as follows:

- Comprehensive worldwide analysis of scientific AI applications and implementations with results (How successful they are in their domain).
- Highlighting the current and future AI challenges and solutions including the ethical view of some latest technologies.
- Future Directions and Policy Recommendations.

This review paper presents the findings of an in-depth investigation conducted on the current level of AI technology. The purpose of this study was to highlight how AI has the potential to alter organizations all over the world. Several challenges were faced while dealing with this

studies like:

- Limited data availability and credibility, particularly in emerging AI applications.
- Constraints related to sample size and diversity affect the generalizability of findings.
- Complexity in synthesizing diverse literature sources and reconciling contradictory findings.
- Time constraints in accessing and analyzing extensive datasets and research materials.
- Difficulty in ensuring the reliability and validity of research methodologies used.
- The collection and integration of visual images posed a notable hurdle. Ensuring the quality, copyright compliance, and effective incorporation of visuals demanded careful attention.
- Challenges in balancing depth and breadth of coverage within the scope of the paper.

These challenges underscore the need for robust research methodologies and interdisciplinary collaboration to effectively address the multifaceted aspects of AI's impact on different industries.

Lastly, while AI offers thrilling opportunities for growth and innovation, it also brings significant ethical, social, and economic concerns. To maximize its benefits and mitigate risks, ongoing dialogue and collaboration are essential. AI's impact on society and industry is profound, and we must approach its research and application with caution, responsibility, and a commitment to ethical considerations. Striving for a future where AI benefits humanity should be our ultimate goal.

CRedit authorship contribution statement

Adib Bin Rashid: Writing – review & editing, Supervision, Conceptualization. **MD Ashfakul Karim Kausik:** Writing – original draft, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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